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# Quantitative Methods I · 215 exam cards

Midterm practice in the format of the paper: five options, **E is always "None of the above"** (and is the right answer on 43 cards).

Each card is followed by its hint and its worked solution. Cover them, work the card with pen, paper and calculator, then check. Every answer was recomputed in Python.

## Domain & range · 17 cards

### 1 Where a square root is defined

What is the domain of $f(x) = \sqrt{2x - 6}$ ?

**A.** $]3, +\infty[$
**B.** $\mathbb{R}$
**C.** $[-3, +\infty[$
**D.** $[6, +\infty[$
**E.** None of the above

**Hint.** A square root needs what is under it to be $\geq 0$ — not $> 0$.

**Answer: E.** $2x - 6 \geq 0 \Leftrightarrow x \geq 3$. Zero is allowed under a root ($\sqrt{0} = 0$), so the bracket is closed: **$[3, +\infty[$**.

> On this card the right answer — **$[3, +\infty[$** — is not among A–D, so the answer is **E · None of the above**.

### 2 Two holes in the line

What is the domain of $f(x) = \dfrac{x + 1}{x^2 - 9}$ ?

**A.** $\mathbb{R} \setminus \{-1, 3\}$
**B.** $\mathbb{R} \setminus \{-3, 3\}$
**C.** $\mathbb{R} \setminus \{3\}$
**D.** $]-3, 3[$
**E.** None of the above

**Hint.** Only the denominator can forbid a value. Where is $x^2 - 9 = 0$?

**Answer: B.** $x^2 - 9 = 0 \Leftrightarrow x = 3$ or $x = -3$ — both roots, not just the positive one. The numerator $x + 1$ forbids nothing. Domain: **$\mathbb{R} \setminus \{-3, 3\}$**.

### 3 A logarithm of a difference

What is the domain of $f(x) = \ln(5 - x)$ ?

**A.** $]-\infty, 5]$
**B.** $]5, +\infty[$
**C.** $\mathbb{R} \setminus \{5\}$
**D.** $]0, 5[$
**E.** None of the above

**Hint.** ln needs a strictly positive argument. Solve $5 - x > 0$.

**Answer: E.** $5 - x > 0 \Leftrightarrow x < 5$. Strict, because $\ln 0$ does not exist: **$]-\infty, 5[$**. Negative x are fine — $\ln(5 - (-10)) = \ln 15$.

> On this card the right answer — **$]-\infty, 5[$** — is not among A–D, so the answer is **E · None of the above**.

## 4 A root over a fraction

What is the domain of $f(x) = \dfrac{\sqrt{x-2}}{x-4}$ ?

**A.** $\mathbb{R} \setminus \{4\}$
**B.** $[2, 4[ \cup ]4, +\infty[$
**C.** $[4, +\infty[$
**D.** $[2, +\infty[$
**E.** None of the above

**Hint.** Two conditions at once: what is under the root ≥ 0, and the denominator ≠ 0.

**Answer: B.** $x - 2 \geq 0$ gives $x \geq 2$; $x - 4 \neq 0$ removes 4. Together: **$[2, 4[ \cup ]4, +\infty[$**. $x = 2$ is allowed: $f(2) = 0 / (-2) = 0$.

## 5 The top of a parabola

What is the range of $f(x) = 3 - x^2$, for $x \in \mathbb{R}$ ?

**A.** $\mathbb{R}$
**B.** $[0, 3]$
**C.** $]-\infty, 3[$
**D.** $]-\infty, 3]$
**E.** None of the above

**Hint.** $x^2$ is never negative, so $3 - x^2$ is never more than … ?

**Answer: D.** $x^2 \geq 0$, so $3 - x^2 \leq 3$, with 3 reached at $x = 0$. As $|x|$ grows, $f$ goes to $-\infty$. Range: **$]-\infty, 3]$** — closed at 3 because it is reached.

## 6 An exponential, lifted

What is the range of $f(x) = e^x + 2$ ?

**A.** $]2, +\infty[$
**B.** $]3, +\infty[$
**C.** $[2, +\infty[$
**D.** $\mathbb{R}$
**E.** None of the above

**Hint.** $e^x$ takes every value in $]0, +\infty[$ and never 0. Add 2 to everything.

**Answer: A.** $e^x \in ]0, +\infty[$, so $e^x + 2 \in$ **$]2, +\infty[$**. 2 is approached as $x \to -\infty$ but never reached, so the bracket is open.

## 7 A root plus one

What is the range of $f(x) = \sqrt{x} + 1$, for $x \geq 0$ ?

**A.** $]1, +\infty[$
**B.** $\mathbb{R}$
**C.** $[0, +\infty[$
**D.** $[-1, +\infty[$
**E.** None of the above

**Hint.** $\sqrt{x}$ starts at 0 (for $x = 0$) and grows without limit.

**Answer: E.** $\sqrt{x} \in [0, +\infty[$, so $\sqrt{x} + 1 \in$ **$[1, +\infty[$**; 1 is reached at $x = 0$.

> On this card the right answer — **$[1, +\infty[$** — is not among A–D, so the answer is **E · None of the above**.

### 8 A price list with limits

A shop sets its price by $p(q) = 40 - 2q$, and only for $0 \le q \le 20$ units. Which prices can occur?

**A.** [0, 40]
**B.** $\mathbb{R}$
**C.** [20, 40]
**D.** [0, 20]
**E.** None of the above

**Hint.** p decreases with q. Compute p at both ends of the allowed q.

**Answer: A.** $p(0) = 40$ and $p(20) = 0$, and p is a straight line in between, so it takes every value from 0 to 40: **[0, 40]**. [0, 20] is the domain, not the range.

### 9 A root in the denominator

What is the domain of $f(x) = \dfrac{1}{\sqrt{8 - 2x}}$ ?

**A.** $]-\infty, 4[$
**B.** $]-\infty, 8[$
**C.** $\mathbb{R} \setminus \{4\}$
**D.** $]-\infty, 4]$
**E.** None of the above

**Hint.** Under the root: $\ge 0$. In the denominator: $\neq 0$. Together: strictly positive.

**Answer: A.** $8 - 2x > 0 \Leftrightarrow x < 4$. 4 itself makes the denominator $\sqrt{0} = 0$, so it is excluded: **$]-\infty, 4[$**.

### 10 A log and a root together

What is the domain of $f(x) = \ln x + \sqrt{3 - x}$ ?

**A.** $]0, +\infty[$
**B.** $]0, 3[$
**C.** $]0, 3]$
**D.** $[0, 3]$
**E.** None of the above

**Hint.** Each piece has its own condition; x must satisfy both.

**Answer: C.** $\ln x$ needs $x > 0$; $\sqrt{(3 - x)}$ needs $x \le 3$. Both: **$]0, 3]$** — open at 0, closed at 3.

### 11 A denominator that is never zero

What is the domain of $f(x) = \dfrac{x - 2}{x^2 + 4}$ ?

**A.** $\mathbb{R} \setminus \{-2, 2\}$
**B.** $\mathbb{R}$
**C.** $]2, +\infty[$
**D.** $\mathbb{R} \setminus \{2\}$
**E.** None of the above

**Hint.** Can $x^2 + 4$ ever be 0?

**Answer: B.** $x^2 \ge 0$, so $x^2 + 4 \ge 4 > 0$ for every x. Nothing is forbidden: $\mathbb{R}$. $x = 2$ only makes f equal to 0, which is allowed.

## 12 A shifted parabola

What is the range of $f(x) = (x - 1)^2 + 5$ ?

**A.** $]5, +\infty[$
**B.** $[0, +\infty[$
**C.** $\mathbb{R}$
**D.** $[5, +\infty[$
**E.** None of the above

**Hint.** A square is ≥ 0, and equals 0 at x = 1.

**Answer: D.** $(x - 1)^2 \geq 0$, so $f \geq 5$, reached at $x = 1$: **$[5, +\infty[$**. 1 is where the minimum is, not its value.

## 13 A negative exponential

What is the range of $f(x) = -2e^x$ ?

**A.** $\mathbb{R}$
**B.** $]-\infty, 0]$
**C.** $]-\infty, 0[$
**D.** $]-\infty, -2]$
**E.** None of the above

**Hint.** $e^x$ takes every value in $]0, +\infty[$. Multiply that whole set by −2.

**Answer: C.** $e^x \in ]0, +\infty[ \Rightarrow -2e^x \in$ **$]-\infty, 0[$**. 0 is never reached; −2 is only the value at x = 0.

## 14 A log of a quotient

What is the domain of $f(x) = \ln\left(\frac{x - 1}{x + 3}\right)$ ?

**A.** $]-\infty, -3] \cup [1, +\infty[$
**B.** $\mathbb{R} \setminus \{-3\}$
**C.** $]-\infty, -3[ \cup ]1, +\infty[$
**D.** $]-3, 1[$
**E.** None of the above

**Hint.** ln needs its argument strictly positive. A quotient is positive when top and bottom have the same sign.

**Answer: C.** Both positive: $x > 1$. Both negative: $x < -3$. So **$]-\infty, -3[ \cup ]1, +\infty[$**. Both ends are open: $x = 1$ gives ln 0, $x = -3$ divides by zero. $]-3, 1[$ is exactly where the quotient is negative.

## 15 An exponential in the denominator

What is the domain of $f(x) = \frac{1}{e^x - 1}$ ?

**A.** $\mathbb{R}$
**B.** $\mathbb{R} \setminus \{0\}$
**C.** $\mathbb{R} \setminus \{e\}$
**D.** $\mathbb{R} \setminus \{1\}$
**E.** None of the above

**Hint.** Only a zero denominator is forbidden. For which x is $e^x = 1$?

**Answer: B.** $e^x = 1 \Leftrightarrow x = 0$. Every other x is allowed — for $x < 0$ the denominator is negative, but not zero. **$\mathbb{R} \setminus \{0\}$**.

### 16 Range of an upward parabola

What is the range of $f(x) = x^2 - 6x + 10$ ?

**A.** $\mathbb{R}$
**B.** $]1, +\infty[$
**C.** $[3, +\infty[$
**D.** $[10, +\infty[$
**E.** None of the above

**Hint.** Complete the square: $x^2 - 6x + 10 = (x - 3)^2 + \dots$ ?

**Answer: E.** $x^2 - 6x + 10 = (x - 3)^2 + 1 \geq 1$, with 1 reached at $x = 3$: **$[1, +\infty[$**. 10 is $f(0)$, and 3 is where the minimum sits, not its value.

> On this card the right answer — **$[1, +\infty[$** — is not among A–D, so the answer is **E · None of the above**.

### 17 A price that settles

After a launch, a product's price follows $p(t) = 20 + 30e^{-0.1t}$ ($t \geq 0$, in months). Which prices can occur?

**A.** $[20, 50]$
**B.** $[30, 50]$
**C.** $]20, 50]$
**D.** $]20, +\infty[$
**E.** None of the above

**Hint.** Compute $p(0)$. Then ask what $e^{-0.1t}$ does as $t$ grows — does it ever reach 0?

**Answer: C.** $p(0) = 20 + 30 = 50$, reached. $e^{-0.1t}$ decreases towards 0 without reaching it, so $p$ approaches 20 but never gets there: **$]20, 50]$**.

## Operations & composition · 16 cards

### 18 A product of two functions

Let $f(x) = 2x + 1$ and $g(x) = x^2$. Compute $(f \cdot g)(3)$.

**A.** 19
**B.** 63
**C.** 16
**D.** 49
**E.** None of the above

**Hint.** $(f \cdot g)(3)$ means $f(3)$ times $g(3)$ — a product, not a composition.

**Answer: B.** $f(3) = 7$, $g(3) = 9$, so $(f \cdot g)(3) = 7 \times 9 =$ **63**. 19 is $f(g(3))$ and 49 is $g(f(3))$ — the two compositions.

### 19 Composition as a formula

Let $f(x) = x^2 - 1$ and $g(x) = x + 3$. What is $(f \circ g)(x)$?

**A.** $x^2 + 2$
**B.** $x^2 + 6x + 8$
**C.** $x^2 + 8$
**D.** $x^2 + 3x + 8$
**E.** None of the above

**Hint.** $f \circ g$ means: first g, then f. Replace every $x$ in $f$ by $(x + 3)$.

**Answer: B.** $(f \circ g)(x) = f(x + 3) = (x + 3)^2 - 1 = x^2 + 6x + 9 - 1 =$ **$x^2 + 6x + 8$**. $x^2 + 2$ is $g \circ f$; $x^2 + 8$ forgets the middle term of $(x + 3)^2$.

### 20 Inside out, with a root

Let $f(x) = \sqrt{x}$ and $g(x) = 4x + 1$. Compute $(g \circ f)(9)$.

**A.** 12
**B.** 37
**C.** 13
**D.** $\sqrt{37}$
**E.** None of the above

**Hint.** First the inner function: f(9). Then feed the result to g.

**Answer: C.** $f(9) = 3$, then $g(3) = 4 \cdot 3 + 1 = \mathbf{13}$. $\sqrt{37}$ is the other order, $f(g(9))$.

### 21 A quotient at a point

Let $f(x) = x^3 - 2$ and $g(x) = x + 2$. Compute $(\frac{f}{g})(2)$.

**A.** $\frac{2}{3}$
**B.** 4
**C.** 1.5
**D.** 6
**E.** None of the above

**Hint.** Evaluate each function at 2 first, then divide.

**Answer: C.** $f(2) = 8 - 2 = 6$, $g(2) = 4$, so $(f/g)(2) = 6/4 = \mathbf{1.5}$. $2/3$ is $g/f$.

### 22 Cost of an hour's production

A workshop produces $q(h) = 12h$ units in h hours, and the cost of q units is $C(q) = 500 + 3q$. Express the cost as a function of the hours worked.

**A.** C = 36h
**B.** C = 536h
**C.** C = 1 500 + 36h
**D.** C = 500 + 12h
**E.** None of the above

**Hint.** Replace q in C(q) by 12h — a composition C(q(h)).

**Answer: E.** $C(q(h)) = 500 + 3(12h) = \mathbf{500 + 36h}$: the fixed 500 does not depend on the hours; each hour adds 12 units at 3 each.

> On this card the right answer — **C = 500 + 36h** — is not among A–D, so the answer is **E · None of the above**.

### 23 The difference f(a + 1) – f(a)

Let $f(x) = x^2$. Simplify $f(a + 1) - f(a)$.

**A.** 1
**B.** 2a + 2
**C.** 2a
**D.** 2a + 1
**E.** None of the above

**Hint.** f(a + 1) means: replace x by the whole bracket (a + 1), then square it.

**Answer: D.** $(a + 1)^2 - a^2 = a^2 + 2a + 1 - a^2 = \mathbf{2a + 1}$. The answer 1 comes from writing f(a + 1) as f(a) + 1 — the classic slip.

## 24 Taking a function apart

$h(x) = (3x - 1)^5$. Which pair gives $h = f \circ g$ ?

**A.** $g(x) = 3x - 1$, $f(u) = 5u$

**B.** $g(x) = 5x$, $f(u) = 3u - 1$

**C.** $g(x) = x^5$, $f(u) = 3u - 1$

**D.** $g(x) = x - 1$, $f(u) = (3u)^5$

**E.** None of the above

**Hint.** g is what you compute first (the inside), f what you do to the result (the outside).

**Answer: E.** Inside: $3x - 1$. Outside: raise to the fifth power. So **$g(x) = 3x - 1$ and $f(u) = u^5$**. The reversed pair gives $3x^5 - 1$; $f(u) = (3u)^5$ applied to $x - 1$ gives $(3x - 3)^5$.

> On this card the right answer — **$g(x) = 3x - 1$, $f(u) = u^5$** — is not among A–D, so the answer is **E · None of the above**.

## 25 A function applied twice

Let $f(x) = 2x - 3$. Compute $f(f(2))$.

**A.** 2

**B.** −5

**C.** 1

**D.** −1

**E.** None of the above

**Hint.** Work from the inside: f(2) first, then f of that.

**Answer: D.** $f(2) = 1$, then $f(1) = 2 - 3 = \mathbf{-1}$. 1 stops after one step; 4 is $(f(2))^2 + 3$, which is not a composition.

## 26 A reciprocal after a shift

Let $f(x) = x + 2$ and $g(x) = \frac{1}{x}$. What is $(g \circ f)(x)$?

**A.** $\frac{x}{x+2}$

**B.** $\frac{1}{2x}$

**C.** $\frac{1}{x} + 2$

**D.** $x + \frac{1}{2}$

**E.** None of the above

**Hint.** $g \circ f$: first add 2, then take the reciprocal.

**Answer: E.** $g(f(x)) = g(x + 2) = \mathbf{1 / (x + 2)}$. $1/x + 2$ is the other order, $f(g(x))$.

> On this card the right answer — $\frac{1}{x+2}$ — is not among A–D, so the answer is **E · None of the above**.

## 27 A difference at a negative point

Let $f(x) = 3x^2$ and $g(x) = x^3 - x$. Compute $(f - g)(-1)$.

**A.** 4

**B.** −3

**C.** 3

**D.** 6

**E.** None of the above

**Hint.** Evaluate each at −1, keeping the signs: $(-1)^2 = 1$, $(-1)^3 = -1$.

**Answer: C.** $f(-1) = 3$, $g(-1) = -1 + 1 = 0$. $(f - g)(-1) = \mathbf{3}$.

### 28 Revenue per hour open

A bakery serves $n(h) = 40 + 5h$ customers when it opens $h$ hours a day, and each customer spends CHF 15. Write the daily revenue as a function of $h$.

**A.** $R = 55 + 75h$
**B.** $R = 75h$
**C.** $R = 40 + 75h$
**D.** $R = 600 + 75h$
**E.** None of the above

**Hint.** $R(n) = 15n$; replace $n$ by $40 + 5h$.

**Answer: D.** $R = 15(40 + 5h) =$ **$600 + 75h$**: the 40 regulars bring 600; each extra hour adds 5 customers × 15.

### 29 Replace x by 2x

Let $f(x) = x^2 + 1$. What is $f(2x)$?

**A.** $2x^2 + 2$
**B.** $2x^2 + 1$
**C.** $4x^2 + 1$
**D.** $4x^2 + 2$
**E.** None of the above

**Hint.** Every $x$ becomes $(2x)$, brackets included.

**Answer: C.** $(2x)^2 + 1 =$ **$4x^2 + 1$**. $2x^2 + 2$ is $2 \cdot f(x)$ — doubling the output, not the input.

### 30 Order matters

Let $f(x) = x + 5$ and $g(x) = 2x$. Compute $(g \circ f)(1) - (f \circ g)(1)$.

**A.** 0
**B.** 5
**C.** −5
**D.** 12
**E.** None of the above

**Hint.** Compute both compositions at 1 separately.

**Answer: B.** $(g \circ f)(1) = g(6) = 12$; $(f \circ g)(1) = f(2) = 7$. Difference: **5**. 0 assumes the order does not matter — it does.

### 31 Solve for the inside

Let $f(x) = 2x + 1$ and $g(x) = x^2$. For which $x > 0$ is $(f \circ g)(x) = 19$?

**A.** $x = \sqrt{19}$
**B.** $x = 3$
**C.** $x = (\sqrt{19} - 1)/2$
**D.** $x = 2$
**E.** None of the above

**Hint.** $(f \circ g)(x) = f(g(x)) = 2x^2 + 1$. Set it equal to 19.

**Answer: B.** $2x^2 + 1 = 19 \Leftrightarrow x^2 = 9 \Leftrightarrow$ **$x = 3$** $(x > 0)$. 9 stops at $x^2$; $(\sqrt{19} - 1)/2$ solves $(g \circ f)(x) = 19$ — the other order.

### 32 Domain of a composition

Let $f(x) = \sqrt{x}$ and $g(x) = 4 - x^2$. What is the domain of $f \circ g$ ?

**A.** $[0, 2]$

**B.** $]-2, 2[$

**C.** $[-4, 4]$

**D.** $\mathbb{R}$

**E.** None of the above

**Hint.** $(f \circ g)(x) = \sqrt{4 - x^2}$. What must be true of $4 - x^2$?

**Answer: E.** $4 - x^2 \geq 0 \Leftrightarrow x^2 \leq 4 \Leftrightarrow$ **$-2 \leq x \leq 2$**. Negative x are fine: $g(-1) = 3$ and $\sqrt{3}$ exists. The ends are included because $\sqrt{0} = 0$.

> On this card the right answer — **[–2, 2]** — is not among A–D, so the answer is **E · None of the above**.

### 33 Shift inside, shift outside

Let $f(x) = x^2$. Write $g(x) = f(x - 3) + 2$ as a polynomial.

**A.** $x^2 - 1$

**B.** $x^2 - 6x + 7$

**C.** $x^2 - 7$

**D.** $x^2 - 6x + 11$

**E.** None of the above

**Hint.** Replace x by $(x - 3)$ INSIDE the square, then add 2 outside.

**Answer: D.** $(x - 3)^2 + 2 = x^2 - 6x + 9 + 2 =$ **$x^2 - 6x + 11$**. $x^2 - 7$ squares each term separately; $x^2 - 1$ subtracts 3 outside instead of inside.

## Inverse functions · 14 cards

### 34 Undo a straight line

Let $f(x) = 4x - 7$. What is $f^{-1}(x)$?

**A.** $\dfrac{1}{4x - 7}$

**B.** $\dfrac{x - 7}{4}$

**C.** $4x + 7$

**D.** $\dfrac{x}{4} + 7$

**E.** None of the above

**Hint.** Write $y = 4x - 7$ and solve for x. Undo the steps in reverse order: first +7, then ÷4.

**Answer: E.** $y = 4x - 7 \Leftrightarrow x = (y + 7)/4$, so **$f^{-1}(x) = (x + 7)/4$**. Check: $f(f^{-1}(5)) = f(3) = 5$. $1/(4x - 7)$ is $1/f$, not $f^{-1}$.

> On this card the right answer — **$\dfrac{x + 7}{4}$** — is not among A–D, so the answer is **E · None of the above**.

### 35 Price → quantity

Demand is $p = 60 - 3q$. Express the quantity demanded as a function of the price.

**A.** $q = 60 - \dfrac{p}{3}$

**B.** $q = \dfrac{p - 60}{3}$

**C.** $q = 20 - \dfrac{p}{3}$

**D.** $q = 20 - 3p$

**E.** None of the above

**Hint.** Solve $p = 60 - 3q$ for q. Move 3q to one side first.

**Answer: C.** $3q = 60 - p \Leftrightarrow q = (60 - p)/3 =$ **$20 - p/3$**. Check: $p = 30$ gives $q = 10$, and $60 - 3 \cdot 10 = 30$.

## 36 Inverse of a rational function

Let $f(x) = \dfrac{2x + 1}{x - 3}$, $x \neq 3$. What is $f^{-1}(x)$?

**A.** $\dfrac{3x - 1}{x - 2}$

**B.** $\dfrac{3x + 1}{x + 2}$

**C.** $\dfrac{3x + 1}{x - 2}$

**D.** $\dfrac{x - 3}{2x + 1}$

**E.** None of the above

**Hint.** $y(x - 3) = 2x + 1$. Expand, collect every x on one side, factor x out.

**Answer: C.** $yx - 3y = 2x + 1 \Leftrightarrow x(y - 2) = 3y + 1 \Leftrightarrow x = (3y + 1)/(y - 2)$. So $\mathbf{f^{-1}(x) = (3x + 1)/(x - 2)}$. $(x - 3)/(2x + 1)$ is $1/f$.

## 37 Undo an exponential

Let $f(x) = e^{2x}$. What is $f^{-1}(x)$, for $x > 0$?

**A.** $\ln x - 2$

**B.** $\dfrac{\ln x}{2}$

**C.** $e^{x/2}$

**D.** $2 \ln x$

**E.** None of the above

**Hint.** $y = e^{2x}$. Take ln of both sides: $\ln y = 2x$.

**Answer: B.** $\ln y = 2x \Leftrightarrow x = \ln(y)/2$, so $\mathbf{f^{-1}(x) = \ln(x)/2}$. Check: $f(\ln(5)/2) = e^{\ln 5} = 5$.

## 38 Undo a shifted logarithm

Let $f(x) = 3 + \ln(x - 1)$, $x > 1$. What is $f^{-1}(x)$?

**A.** $e^{x - 3} - 1$

**B.** $1 + e^{x - 3}$

**C.** $1 + e^{x + 3}$

**D.** $\ln(x - 3) + 1$

**E.** None of the above

**Hint.** Undo in reverse order: subtract 3, then exponentiate, then add 1.

**Answer: B.** $y - 3 = \ln(x - 1) \Leftrightarrow e^{y - 3} = x - 1 \Leftrightarrow x = 1 + e^{y - 3}$. So $\mathbf{f^{-1}(x) = 1 + e^{x - 3}}$.

## 39 Undo a cube

Let $f(x) = x^3 + 2$. What is $f^{-1}(x)$?

**A.** $\sqrt[3]{(x + 2)}$

**B.** $(x - 2)^3$

**C.** $\sqrt[3]{(x - 2)}$

**D.** $\dfrac{1}{x^3 + 2}$

**E.** None of the above

**Hint.** Subtract 2 first, then take the cube root — the reverse of 'cube, then add 2'.

**Answer: C.** $y = x^3 + 2 \Leftrightarrow x^3 = y - 2 \Leftrightarrow x = \sqrt[3]{(y - 2)}$. $\mathbf{f^{-1}(x) = \sqrt[3]{(x - 2)}}$. $\sqrt[3]{x} - 2$ undoes the steps in the wrong order.

## 40 Undo a division

Let $f(x) = \dfrac{x-4}{3}$. What is $f^{-1}(x)$?

**A.** $3x + 4$

**B.** $3x - 4$

**C.** $\dfrac{3}{x-4}$

**D.** $\dfrac{x+4}{3}$

**E.** None of the above

**Hint.** f subtracts 4 then divides by 3. Undo: multiply by 3, then add 4.

**Answer: A.** $y = (x-4)/3 \Leftrightarrow 3y = x - 4 \Leftrightarrow x = 3y + 4$: **$f^{-1}(x) = 3x + 4$**.

## 41 Supply, read the other way

Supply is $p = 2q + 10$. Express the quantity supplied as a function of the price.

**A.** $q = 2p - 10$

**B.** $q = \dfrac{p}{2} - 10$

**C.** $q = \dfrac{p+10}{2}$

**D.** $q = 2(p - 10)$

**E.** None of the above

**Hint.** Solve $p = 2q + 10$ for q.

**Answer: E.** $2q = p - 10 \Leftrightarrow$ **$q = (p-10)/2$**. At $p = 10$ nothing is supplied; each extra franc brings half a unit.

> On this card the right answer — **$q = \dfrac{p-10}{2}$** — is not among A–D, so the answer is **E · None of the above**.

## 42 Undo a reciprocal

Let $f(x) = \dfrac{5}{x+1}$, $x \neq -1$. What is $f^{-1}(x)$?

**A.** $\dfrac{5}{x} - 1$

**B.** $\dfrac{5}{x} + 1$

**C.** $\dfrac{5}{x-1}$

**D.** $\dfrac{x+1}{5}$

**E.** None of the above

**Hint.** $y(x+1) = 5 \Rightarrow x + 1 = 5/y$.

**Answer: A.** $x + 1 = 5/y \Leftrightarrow x = 5/y - 1$: **$f^{-1}(x) = 5/x - 1$**. $(x+1)/5$ is $1/f$.

## 43 Undo ln(2x)

Let $f(x) = \ln(2x)$, $x > 0$. What is $f^{-1}(x)$?

**A.** $\dfrac{e^x}{2}$

**B.** $2e^x$

**C.** $e^{x/2}$

**D.** $\dfrac{\ln x}{2}$

**E.** None of the above

**Hint.** $y = \ln(2x) \Leftrightarrow e^y = 2x$.

**Answer: A.** $e^y = 2x \Leftrightarrow x = e^y/2$: **$f^{-1}(x) = e^x/2$**.

## 44 Undo a root

Let $f(x) = \sqrt{x + 4}$, $x \geq -4$. What is $f^{-1}(x)$, for $x \geq 0$?

**A.** $\sqrt{x - 4}$

**B.** $\sqrt{x} - 4$

**C.** $x^2 + 4$

**D.** $(x - 4)^2$

**E.** None of the above

**Hint.** Square both sides of $y = \sqrt{(x + 4)}$.

**Answer: E.** $y^2 = x + 4 \Leftrightarrow x = y^2 - 4$: **$f^{-1}(x) = x^2 - 4$**. Check: $f(5) = 3$ and $3^2 - 4 = 5$ ✓.

> On this card the right answer — **$x^2 - 4$** — is not among A–D, so the answer is **E · None of the above**.

## 45 Undo an exponential with a minus

Let $f(x) = 5 - 2e^x$. What is $f^{-1}(x)$, for $x < 5$?

**A.** $\ln(\frac{x - 5}{2})$

**B.** $\frac{\ln(5 - x)}{2}$

**C.** $\ln(5 - 2x)$

**D.** $\ln(\frac{5 - x}{2})$

**E.** None of the above

**Hint.** $y = 5 - 2e^x$. Isolate $e^x$ first, then take ln.

**Answer: D.** $2e^x = 5 - y \Leftrightarrow e^x = (5 - y)/2 \Leftrightarrow x = \ln((5 - y)/2)$. So **$f^{-1}(x) = \ln((5 - x)/2)$**, defined for $x < 5$ — exactly the range of f. $\ln(5 - x)/2$ divides the log instead of its argument.

## 46 When will sales reach a level?

Sales grow as $f(x) = 100 \cdot 1.04^x$ (x in years). Which formula gives the year x at which sales reach a level s?

**A.** $\frac{\ln(s/100)}{\ln 1.04}$

**B.** $\frac{\ln s}{100 \ln 1.04}$

**C.** $\frac{\ln(s - 100)}{\ln 1.04}$

**D.** $\frac{\ln(s/100)}{1.04}$

**E.** None of the above

**Hint.** Solve $s = 100 \cdot 1.04^x$ for x: divide by 100 first, then take ln of both sides.

**Answer: A.** $1.04^x = s/100 \Leftrightarrow x \ln 1.04 = \ln(s/100) \Leftrightarrow$ **$x = \ln(s/100)/\ln 1.04$**. For $s = 200$: $\ln 2/\ln 1.04 \approx 17.7$ years. Dividing 100 out must happen before the log, not inside it.

## 47 A value of the inverse, without the formula

Let $f(x) = x^3 + x + 1$, which is strictly increasing. What is $f^{-1}(3)$?

**A.** 2

**B.** 1

**C.** $\sqrt[3]{2}$

**D.** $\frac{1}{31}$

**E.** None of the above

**Hint.** $f^{-1}(3)$ is the x for which $f(x) = 3$. Try small whole numbers.

**Answer: B.** $f(1) = 1 + 1 + 1 = 3$, so **$f^{-1}(3) = 1$**; f is strictly increasing ($f' = 3x^2 + 1 > 0$), so no other x works. 31 is $f(3)$, and $1/31$ is $1/f(3)$ — neither is the inverse.

## Exponentials & logarithms · 18 cards

### 48 Solve an exponential equation

Solve $e^{2x} = 9$.

**A.** $x = \ln 3$

**B.** $x = 3$

**C.** $x = \ln 9$

**D.** $x = \ln 4.5$

**E.** None of the above

**Hint.** Take ln of both sides; $\ln(e^{2x}) = 2x$.

**Answer: A.** $2x = \ln 9 = 2 \ln 3$, so **$x = \ln 3 \approx 1.099$**. $\ln 4.5$ divides inside the log, which is not allowed.

### 49 Logs in other bases

Compute $\log_2(32) + \log_3(\frac{1}{9})$.

**A.** 4

**B.** –3

**C.** 1.5

**D.** 3

**E.** None of the above

**Hint.** $\log_2 32$ asks: 2 to what power is 32? $\log_3(1/9)$: 3 to what power is 1/9?

**Answer: D.** $2^5 = 32 \rightarrow 5$. $3^{-2} = 1/9 \rightarrow -2$. Sum: **3**. 7 forgets the minus sign of the second log.

### 50 A power of e inside ln

Simplify $\ln(e^3 \cdot \sqrt{e})$.

**A.** $e^{3.5}$

**B.** 1.5

**C.** 4

**D.** 3.5

**E.** None of the above

**Hint.** $\sqrt{e} = e^{1/2}$; multiplying powers adds the exponents; $\ln(e^k) = k$.

**Answer: D.** $e^3 \cdot e^{1/2} = e^{3.5}$, and $\ln(e^{3.5}) =$ **3.5**.

### 51 Expand a logarithm

Write $\ln(\frac{x^2 y}{z^3})$ in terms of ln x, ln y, ln z ($x, y, z > 0$).

**A.** $2 \ln x \cdot \ln y - 3 \ln z$

**B.** $\frac{2 \ln x + \ln y}{3 \ln z}$

**C.** $2 \ln x + \ln y - 3 \ln z$

**D.** $\ln 2x + \ln y - \ln 3z$

**E.** None of the above

**Hint.** Product → sum, quotient → difference, power → a factor in front.

**Answer: C.** $\ln(x^2) + \ln(y) - \ln(z^3) =$ **$2 \ln x + \ln y - 3 \ln z$**. A quotient inside ln becomes a subtraction, never a division of logs.

## 52 Combine into one logarithm

Write $2 \ln a - \ln b + \frac{1}{2} \ln c$ as a single logarithm ($a, b, c > 0$).

**A.** $\ln(a^2 - b + \sqrt{c})$

**B.** $\ln(\frac{a^2\sqrt{c}}{b})$

**C.** $\ln(a^2 b\sqrt{c})$

**D.** $\ln(\frac{a^2 c}{2b})$

**E.** None of the above

**Hint.** A factor in front becomes a power inside; + becomes ×, − becomes ÷.

**Answer: B.** $2 \ln a = \ln a^2$, $\frac{1}{2} \ln c = \ln \sqrt{c}$. Then $\ln a^2 - \ln b + \ln \sqrt{c} = \mathbf{\ln(a^2\sqrt{c}/b)}$.

## 53 Solve with a base-2 power

Solve $5 \cdot 2^x = 40$.

**A.** $x = 2$

**B.** $x = 4$

**C.** $x = \ln 8$

**D.** $x = 3$

**E.** None of the above

**Hint.** Divide by 5 first, then ask: 2 to what power is 8?

**Answer: D.** $2^x = 8 = 2^3$, so **$x = 3$**. With logs: $x = \ln 8 / \ln 2 = 3$. $x = 8$ stops one step early.

## 54 How long until it doubles

CHF 1 000 grows at 4% a year, compounded continuously: $V(t) = 1\,000\, e^{0.04t}$. After how many years does it reach CHF 2 000?

**A.** 25 years

**B.** 50 years

**C.** ≈ 17.33 years

**D.** 12.5 years

**E.** None of the above

**Hint.** Set $V(t) = 2\,000$, divide by 1 000, take ln.

**Answer: C.** $e^{0.04t} = 2 \Leftrightarrow 0.04t = \ln 2 \Leftrightarrow t = 0.6931 / 0.04 =$ **17.33 years**. 17.67 is the answer for yearly compounding ($\ln 2 / \ln 1.04$) — a different model.

## 55 Two logs equal

Solve $\ln(x + 2) = \ln(3x - 4)$.

**A.** $x = 6$

**B.** $x = -3$

**C.** $x = 1$

**D.** $x = 3$

**E.** None of the above

**Hint.** ln is one-to-one: equal logs mean equal arguments. Then check both arguments are positive.

**Answer: D.** $x + 2 = 3x - 4 \Leftrightarrow 2x = 6 \Leftrightarrow$ **$x = 3$**. Check: $\ln 5 = \ln 5$, both arguments positive.

### 56 A small number in base 10

Compute $\log_{10}(0.001) + \ln 1$.

**A.** 3
**B.** –3
**C.** –4
**D.** –2
**E.** None of the above

**Hint.** 0.001 = 10 to what power? And ln 1 = ?

**Answer: B.** $0.001 = 10^{-3} \rightarrow -3$. $\ln 1 = 0$. Sum: **–3**.

### 57 Undo a natural log

Solve $\ln x = 2$.

**A.** $x = e^2 \approx 7.389$
**B.** $x = \sqrt{e}$
**C.** $x = 100$
**D.** $x = \ln 2$
**E.** None of the above

**Hint.** $\ln x = 2$ means: e to the power 2 is x.

**Answer: A.** $x = \mathbf{e^2 \approx 7.389}$. 100 would be the answer for $\log_{10} x = 2$.

### 58 Powers of three

Solve $3^{x+1} = 81$.

**A.** x = 27
**B.** x = 4
**C.** x = 2
**D.** x = 26
**E.** None of the above

**Hint.** $81 = 3^4$. Compare the exponents.

**Answer: E.** $3^{x+1} = 3^4 \Leftrightarrow x + 1 = 4 \Leftrightarrow \mathbf{x = 3}$. 4 forgets to subtract 1.

> On this card the right answer — **x = 3** — is not among A–D, so the answer is **E · None of the above**.

### 59 ln of e and of 1⁄e

Compute $\ln(e^5) - \ln(\frac{1}{e})$.

**A.** 5
**B.** 4
**C.** –6
**D.** $e^6$
**E.** None of the above

**Hint.** $\ln(e^k) = k$, and $1/e = e^{-1}$.

**Answer: E.** $5 - (-1) = \mathbf{6}$. 4 drops the minus sign of $\ln(1/e)$.

> On this card the right answer — **6** — is not among A–D, so the answer is **E · None of the above**.

### 60 Which one equals ln 12?

Which expression is equal to ln 12?

**A.** 2 ln 6
**B.** ln 3 · ln 4
**C.** ln 24 − ln 4
**D.** ln 3 + 2 ln 2
**E.** None of the above

**Hint.** $12 = 3 \times 4 = 3 \times 2^2$. Use $\ln(ab) = \ln a + \ln b$ and $\ln(a^k) = k \ln a$.

**Answer: D.** $\ln 12 = \ln 3 + \ln 4 =$ **$\ln 3 + 2 \ln 2$**. $2 \ln 6 = \ln 36$; $\ln 24 - \ln 4 = \ln 6$; $\ln 3 \cdot \ln 4$ is not a log law at all.

### 61 Sales that triple

Monthly sales follow $S(t) = 500\, e^{0.1t}$ (t in months). When do they reach 1 500?

**A.** $t = 10 \ln 3 \approx 11$ months
**B.** 20 months
**C.** 10 months
**D.** 30 months
**E.** None of the above

**Hint.** Divide by 500, then take ln.

**Answer: A.** $e^{0.1t} = 3 \Leftrightarrow 0.1t = \ln 3 \Leftrightarrow$ **$t = 10 \ln 3 \approx 10.99$**. 30 treats the growth as linear.

### 62 Half the value

A car's value is $V(t) = 8\,000\, e^{-0.25t}$ (t in years). After how long is it worth half?

**A.** 2 years
**B.** 4 years
**C.** ≈ 0.17 years
**D.** ≈ 2.77 years
**E.** None of the above

**Hint.** Set $e^{-0.25t} = ½$ and take ln.

**Answer: D.** $-0.25t = \ln ½ = -\ln 2 \Leftrightarrow t = \ln 2 / 0.25 =$ **2.77 years**. 0.17 multiplies by 0.25 instead of dividing.

### 63 A sum of logs

Solve $\ln(x + 2) + \ln(x - 2) = \ln 5$.

**A.** $x = \sqrt{5}$
**B.** $x = 3$ or $x = -3$
**C.** $x = 3$
**D.** $x = 9$
**E.** None of the above

**Hint.** $\ln a + \ln b = \ln(ab)$. Then check that every log you started with exists.

**Answer: C.** $\ln((x + 2)(x - 2)) = \ln 5 \Leftrightarrow x^2 - 4 = 5 \Leftrightarrow x = \pm 3$. $x = -3$ makes both $x + 2$ and $x - 2$ negative, so their logs do not exist: **$x = 3$** only.

### 64 e to a combination of logs

Compute $e^{2 \ln 3 - \ln 6}$ without a calculator.

**A.** 54
**B.** 0.5
**C.** 1.5
**D.** 3
**E.** None of the above

**Hint.** First turn 2 ln 3 – ln 6 into ONE logarithm.

**Answer: C.** 2 ln 3 – ln 6 = ln 9 – ln 6 = ln(9/6) = ln 1.5, and $e^{\ln 1.5}$ = **1.5**. 54 = 9 × 6 adds the logs instead of subtracting them.

### 65 A continuous growth rate

Revenue follows $R(t) = 200e^{kt}$ and grows from 200 to 300 in 4 years. What is k?

**A.** 0.125
**B.** ln 1.5 ≈ 0.405
**C.** $\log_{10} 1.5/4 \approx 0.044$
**D.** 0.5
**E.** None of the above

**Hint.** $300 = 200e^{4k}$. Divide, then take ln.

**Answer: E.** $e^{4k} = 1.5 \Leftrightarrow 4k = \ln 1.5 \Leftrightarrow$ **k = ln 1.5/4 ≈ 0.101**, about 10.1% a year, continuously. 0.125 = 50%/4 ignores compounding; ln 1.5 forgets to divide by the 4 years.

> On this card the right answer — **ln 1.5/4 ≈ 0.101** — is not among A–D, so the answer is **E · None of the above**.

## Lines · 19 cards

### 66 A line through two points

Which line passes through (–2, 1) and (4, 13)?

**A.** y = 6x + 13
**B.** y = 2x + 5
**C.** y = 2x + 1
**D.** y = 2x – 3
**E.** None of the above

**Hint.** Slope = rise/run between the two points. Then use one point to find the intercept.

**Answer: B.** m = (13 – 1)/(4 – (–2)) = 12/6 = 2. 1 = 2(–2) + b ⇔ b = 5. **y = 2x + 5**; check (4, 13): 8 + 5 = 13 ✓.

### 67 Where a line crosses the axis

Where does the line 4x + 3y = 24 cross the x-axis?

**A.** (6, 0)
**B.** (4, 0)
**C.** (0, 8)
**D.** (0, 6)
**E.** None of the above

**Hint.** On the x-axis, y = 0.

**Answer: A.** y = 0 ⇒ 4x = 24 ⇒ x = 6: **(6, 0)**. (0, 8) is where it crosses the y-axis.

## 68 Parallel through a point

Which line is parallel to $y = -3x + 1$ and passes through (2, 4)?

**A.** $y = -3x - 2$

**B.** $y = 3x - 2$

**C.** $y = -3x + 10$

**D.** $y = \frac{1}{3}x + \frac{10}{3}$

**E.** None of the above

**Hint.** Parallel lines have the SAME slope. Only the intercept changes.

**Answer: C.** Slope –3. $4 = -3(2) + b \Leftrightarrow b = 10$: **$y = -3x + 10$**. The slope 1⁄3 would be for a perpendicular, and with a sign error.

## 69 Read a slope from the general form

What is the slope of the line $6x - 2y + 7 = 0$?

**A.** 3

**B.** $\frac{7}{2}$

**C.** 6

**D.** $\frac{1}{3}$

**E.** None of the above

**Hint.** Solve for y first: put the equation in the form $y = mx + b$.

**Answer: A.** $2y = 6x + 7 \Leftrightarrow y = 3x + 3.5$. Slope **3**. –3 comes from forgetting that –2y moves across with its sign.

## 70 A taxi fare

A taxi charges a fixed fee plus a price per km. A 12 km trip costs CHF 31, a 20 km trip costs CHF 47. What is the fixed fee?

**A.** CHF 31

**B.** CHF 12

**C.** CHF 15

**D.** CHF 2

**E.** None of the above

**Hint.** The 8 extra km cost 16 francs: that gives the price per km. The fee is what is left of 31.

**Answer: E.** Per km: $(47 - 31)/(20 - 12) = 2$. Fee: $31 - 12 \times 2$ = **CHF 7**. The fare is $C = 7 + 2x$.

> On this card the right answer — **CHF 7** — is not among A–D, so the answer is **E · None of the above**.

## 71 Where two lines meet

Where do $y = 2x - 1$ and $y = -x + 8$ intersect?

**A.** (3, –5)

**B.** (–3, 5)

**C.** (5, 3)

**D.** (7, 13)

**E.** None of the above

**Hint.** At the intersection both y's are equal: $2x - 1 = -x + 8$.

**Answer: E.** $3x = 9 \Leftrightarrow x = 3$, $y = 2 \cdot 3 - 1 = 5$: **(3, 5)**. Check in the second line: $-3 + 8 = 5$ ✓.

> On this card the right answer — **(3, 5)** — is not among A–D, so the answer is **E · None of the above**.

### 72 Which point is on the line

Which point lies on $y = -\frac{1}{2}x + 4$ ?

**A.** (–2, 3)
**B.** (4, 4)
**C.** (6, 1)
**D.** (2, 5)
**E.** None of the above

**Hint.** Put each x into the equation and compare with the y.

**Answer: C.** x = 6: –3 + 4 = 1 ✓ → **(6, 1)**. x = 2 gives 3, x = 4 gives 2, x = –2 gives 5, x = 8 gives 0.

### 73 Straight-line depreciation

A machine bought for CHF 18 000 is worth CHF 3 000 after 5 years, losing value at a constant rate. What is its value V after t years?

**A.** V = 18 000 – 15 000t
**B.** V = 15 000 – 3 000t
**C.** V = 18 000 – 3 000t
**D.** V = 3 000 + 18 000t
**E.** None of the above

**Hint.** Two points: (0, 18 000) and (5, 3 000). The slope is the loss per year.

**Answer: C.** Slope (3 000 – 18 000)/5 = –3 000 a year; intercept 18 000: **V = 18 000 – 3 000t**. Check t = 5: 3 000 ✓.

### 74 From slope and intercept to the general form

A line has slope –4 and y-intercept 7. Which equation describes it?

**A.** x + 4y – 7 = 0
**B.** 4x + y – 7 = 0
**C.** 4x + y – 4 = 0
**D.** 4x + y + 7 = 0
**E.** None of the above

**Hint.** Solve each option for y and compare with y = –4x + 7.

**Answer: B.** 4x + y – 7 = 0 ⇔ y = –4x + 7 ✓ → **4x + y – 7 = 0**. 4x – y + 7 = 0 gives y = 4x + 7 — the right intercept, the wrong slope.

### 75 The intercept from one point

A line has slope –3 and passes through (2, 7). What is its y-intercept?

**A.** 7
**B.** 4
**C.** 1
**D.** –13
**E.** None of the above

**Hint.** y = –3x + b; put in (2, 7).

**Answer: E.** 7 = –6 + b ⇔ **b = 13**. 1 comes from 7 – 6, a sign slip.

> On this card the right answer — **13** — is not among A–D, so the answer is **E · None of the above**.

### 76 Two points with the same x

Which equation describes the line through (1, 4) and (1, –2)?

**A.** $y = 1$
**B.** $y = x + 3$
**C.** $y = -3x + 7$
**D.** $x = 1$
**E.** None of the above

**Hint.** Both points have x = 1. What does a line through them look like?

**Answer: D.** Same x, different y: the line is vertical, **x = 1**. Its slope (6⁄0) does not exist, so no y = mx + b form fits.

### 77 A perpendicular slope

What is the slope of any line perpendicular to $y = \frac{2}{5}x - 1$ ?

**A.** $-\frac{5}{2}$
**B.** $-1$
**C.** $\frac{5}{2}$
**D.** $-\frac{2}{5}$
**E.** None of the above

**Hint.** Perpendicular slopes multiply to –1: flip the fraction and change the sign.

**Answer: A.** $m_2 = -1 / (2/5) =$ **$-5/2$**. Check: $(2/5)(-5/2) = -1$ ✓.

### 78 From the two intercepts

A line crosses the x-axis at 4 and the y-axis at –2. What is its equation?

**A.** $y = 2x - 4$
**B.** $y = \frac{x}{2} - 2$
**C.** $y = -2x + 4$
**D.** $y = \frac{x}{2} + 4$
**E.** None of the above

**Hint.** Two points: (4, 0) and (0, –2).

**Answer: B.** m = (0 – (–2)) ⁄ (4 – 0) = ½, b = –2: **$y = x/2 - 2$**. Check x = 4: 2 – 2 = 0 ✓.

### 79 Two phone plans

Plan A: CHF 20 a month plus CHF 0.10 a minute. Plan B: CHF 35 a month, unlimited. Above how many minutes is B cheaper?

**A.** 15 minutes
**B.** 150 minutes
**C.** 350 minutes
**D.** 55 minutes
**E.** None of the above

**Hint.** Set the two costs equal: 20 + 0.10m = 35.

**Answer: B.** 0.10m = 15 ⇔ **m = 150**. Above 150 minutes plan A costs more than 35.

### 80 Read a value off a line

A line passes through (0, 0) and (3, −6). What is y at x = −2?

**A.** 6
**B.** −4
**C.** 2
**D.** 4
**E.** None of the above

**Hint.** Slope first, then y = mx (the line passes through the origin).

**Answer: D.** m = −6 / 3 = −2, so y = −2x and y(−2) = **4**.

### 81 A line from a table

A linear relation gives y = 11 at x = 2, y = 8 at x = 4 and y = 5 at x = 6. What is its equation?

**A.** y = −3x + 17
**B.** y = 1.5x + 8
**C.** $y = -\frac{2}{3}x + \frac{37}{3}$
**D.** y = −1.5x + 14
**E.** None of the above

**Hint.** Slope = Δy / Δx between any two rows. The intercept is y at x = 0, not y at the first row.

**Answer: D.** m = (8 − 11) / (4 − 2) = −1.5. b = 11 − (−1.5)(2) = 14: **y = −1.5x + 14**. −3 takes Δy for a step of 2 as if it were a step of 1; −2⁄3 is Δx / Δy.

### 82 A sales trend

A firm sold 1 200 units in 2020 and 1 560 in 2026, and sales grow linearly. What does the trend predict for 2030?

**A.** 1 900
**B.** 1 680
**C.** 1 800
**D.** 1 720
**E.** None of the above

**Hint.** Slope = extra units per year = 360 / 6. Then add 4 more years to 2026.

**Answer: C.** m = (1 560 − 1 200) / (2026 − 2020) = 60 units a year. 2030: 1 560 + 4 × 60 = **1 800**. 2 040 uses 360 / 3; 1 680 adds only 2 years.

### 83 Which pair is perpendicular?

Which pair of lines is perpendicular?

**A.** y = 2x + 1 and y = −½x + 4
**B.** y = 2x + 1 and y = 2x − 3
**C.** y = −2x + 1 and y = −½x + 3
**D.** y = 4x and y = ¼x + 1
**E.** None of the above

**Hint.** Perpendicular slopes multiply to −1: $m_2 = -1 / m_1$ — the reciprocal AND the opposite sign.

**Answer: A.** 2 × (−½) = −1: **y = 2x + 1 and y = −½x + 4**. Equal slopes are parallel; 3 and −3 only flip the sign; 4 and ¼ only invert; −2 × −½ = +1.

### 84 A line with slope zero

What is the equation of the line through (–2, 3) with slope 0?

**A.** y = x + 5
**B.** y = 3x
**C.** x = –2
**D.** y = –2
**E.** None of the above

**Hint.** Slope 0 means y never changes as x moves.

**Answer: E.** y – 3 = 0 · (x + 2) ⇔ **y = 3**, a horizontal line. x = –2 is the vertical line through the point — its slope is undefined, not 0.

> On this card the right answer — **y = 3** — is not among A–D, so the answer is **E · None of the above**.

## Business models · 26 cards

### 85 A food truck breaks even

A food truck pays CHF 2 400 a month in rent and licences. Each meal sells for CHF 25 and costs CHF 13 in ingredients. How many meals a month cover all costs?

**A.** 96
**B.** 200
**C.** ≈ 184.6
**D.** ≈ 63.2
**E.** None of the above

**Hint.** Each meal contributes 25 – 13 to the fixed costs. How many contributions make 2 400?

**Answer: B.** Break-even: 25q = 2 400 + 13q ⇔ 12q = 2 400 ⇔ **q = 200** meals. 96 divides by the price only, ignoring ingredients.

### 86 Profit at a given volume

Cost C(q) = 1 500 + 8q, revenue R(q) = 14q. What is the profit at q = 400?

**A.** CHF 5 600
**B.** CHF 900
**C.** CHF 2 100
**D.** –CHF 900
**E.** None of the above

**Hint.** Profit = revenue – cost, both at q = 400.

**Answer: B.** R = 5 600, C = 1 500 + 3 200 = 4 700. P = **900**. 2 100 forgets the fixed cost.

### 87 Equilibrium, quantity as a function of price

Demand: q = 120 – 4p. Supply: q = 6p – 30. Find the equilibrium (p, q).

**A.** (60, 15)
**B.** (18, 48)
**C.** (9, 84)
**D.** (12, 72)
**E.** None of the above

**Hint.** Here quantity is written as a function of price, so set the two q's equal and solve for p.

**Answer: E.** 120 – 4p = 6p – 30 ⇔ 10p = 150 ⇔ p = 15, q = 120 – 60 = 60: **(p, q) = (15, 60)**. Check supply: 90 – 30 = 60 ✓.

> On this card the right answer — **(15, 60)** — is not among A–D, so the answer is **E · None of the above**.

### 88 Demand from two observations

At a price of CHF 8 a café sells 300 cups a day; at CHF 10 it sells 260. Assuming a linear demand, what is q as a function of p?

**A.** q = 140 + 20p
**B.** q = 460 + 20p
**C.** q = 460 – 10p
**D.** q = 460 – 20p
**E.** None of the above

**Hint.** Slope = change in q / change in p. Then use one observation for the intercept.

**Answer: D.** Slope (260 – 300) / (10 – 8) = –20. 300 = b – 20·8 ⇔ b = 460: **q = 460 – 20p**. Check p = 10: 260 ✓.

### 89 The revenue-maximising price

Demand is q = 300 – 5p. Which price maximises revenue R = p · q?

**A.** p = 30
**B.** p = 45
**C.** p = 60
**D.** p = 15
**E.** None of the above

**Hint.** R(p) = 300p – 5p² is a downward parabola; its top is halfway between its two zeros.

**Answer: A.** R(p) = p(300 – 5p) is zero at p = 0 and p = 60; the vertex is halfway: **p = 30** (R = 4 500). 60 is where nobody buys.

### 90 Build the profit function

Demand p = 50 – 0.5q, cost C(q) = 200 + 10q. What is the profit P(q)?

**A.** –0.5q² + 40q – 200
**B.** –0.5q² + 50q – 210
**C.** –0.5q² + 40q + 200
**D.** –0.5q² + 60q – 200
**E.** None of the above

**Hint.** Revenue is p times q, with p replaced by the demand formula. Then subtract the WHOLE cost.

**Answer: A.** R = (50 – 0.5q)q = 50q – 0.5q². P = 50q – 0.5q² – 200 – 10q = **–0.5q² + 40q – 200**. 60q adds the unit cost instead of subtracting it.

### 91 The best profit a parabola allows

P(q) = –2q² + 96q – 700. What is the maximum profit?

**A.** 1 152
**B.** 1 604
**C.** 604
**D.** 24
**E.** None of the above

**Hint.** The vertex of a·q² + b·q + c is at q = –b / (2a). The question asks for the PROFIT there, not the quantity.

**Answer: E.** q* = –96 / (–4) = 24. P(24) = –1 152 + 2 304 – 700 = **452**. 24 is the quantity, not the profit.

> On this card the right answer — **452** — is not among A–D, so the answer is **E · None of the above**.

### 92 Two products, two constraints

A bakery makes croissants (c) and baguettes (b). Labour: 2 min per croissant, 3 min per baguette, 480 min available. Butter: 50 g per croissant, 10 g per baguette, 6.8 kg available. How many of each use up both exactly?

**A.** c = 120, b = 60
**B.** c = 96, b = 96
**C.** c = 80, b = 120
**D.** c = 120, b = 80
**E.** None of the above

**Hint.** Two equations: 2c + 3b = 480 and 50c + 10b = 6 800 (grams!). Solve one for b and substitute.

**Answer: D.** From the second: b = 680 − 5c. Then 2c + 3(680 − 5c) = 480 ⇔ −13c = −1 560 ⇔ c = 120, b = 80. **120 croissants, 80 baguettes**. Check: 240 + 240 = 480 ✓; 6 000 + 800 = 6 800 g ✓.

### 93 Fixed cost from two production runs

Producing 100 units costs CHF 3 200; producing 250 units costs CHF 5 600. If cost is linear in output, what is the fixed cost?

**A.** CHF 16
**B.** CHF 1 600
**C.** CHF 2 400
**D.** CHF 1 000
**E.** None of the above

**Hint.** The 150 extra units cost 2 400: that is the variable cost per unit. Fixed = cost at 0 units.

**Answer: B.** Per unit: 2 400 / 150 = 16. Fixed: 3 200 − 100 × 16 = **CHF 1 600**. 16 is the marginal cost.

### 94 Average cost at a volume

C(q) = 900 + 6q. What is the average cost per unit at q = 150?

**A.** CHF 906
**B.** CHF 1 800
**C.** CHF 12
**D.** CHF 6
**E.** None of the above

**Hint.** Average cost = total cost / quantity.

**Answer: C.** C(150) = 900 + 900 = 1 800; 1 800 / 150 = **CHF 12**. 6 is the marginal cost; 1 800 is the total.

### 95 A per-unit tax moves the market

Demand p = 100 − 2q, supply p = 20 + 2q. A tax of CHF 8 per unit is added to the supply price. What price do buyers pay at the new equilibrium?

**A.** CHF 68
**B.** CHF 64
**C.** CHF 60
**D.** CHF 72
**E.** None of the above

**Hint.** With the tax, supply becomes p = 28 + 2q. Set it equal to demand.

**Answer: B.** 100 − 2q = 28 + 2q ⇔ q = 18, p = 100 − 36 = **64**. Without tax p = 60 — buyers carry 4 of the 8 francs. 68 assumes they carry all of it.

### 96 Up 25%, then down 20%

A price rises by 25%, then falls by 20%. What is the overall change?

**A.** +5%
**B.** +45%
**C.** +2.5%
**D.** –5%
**E.** None of the above

**Hint.** Percentages multiply, they do not add: × 1.25, then × 0.80.

**Answer: E.** 1.25 × 0.80 = 1.00 → **0%**: back exactly where it started. +5% adds the percentages, which is the trap.

> On this card the right answer — **0%** — is not among A–D, so the answer is **E · None of the above**.

### 97 A profit target

A product sells for CHF 40, costs CHF 28 per unit to make, and the fixed costs are CHF 9 000. How many units give a profit of CHF 3 000?

**A.** 750
**B.** 300
**C.** 1 250
**D.** ≈ 428.6
**E.** None of the above

**Hint.** Profit = (40 – 28)q – 9 000. Set it equal to 3 000.

**Answer: E.** 12q = 12 000 ⇔ **q = 1 000**. 750 is break-even (profit 0).

> On this card the right answer — **1 000** — is not among A–D, so the answer is **E · None of the above**.

### 98 Equilibrium quantity

Demand q = 500 – 10p, supply q = 40p. What quantity is traded at equilibrium?

**A.** 10
**B.** 450
**C.** 400
**D.** 500
**E.** None of the above

**Hint.** Set demand = supply, solve for p, then compute q.

**Answer: C.** 500 – 10p = 40p ⇔ p = 10, q = 40 × 10 = **400**. 10 is the price, not the quantity.

### 99 Revenue at a price

Demand is q = 80 – 2p. What is the revenue at p = 15?

**A.** CHF 600
**B.** CHF 50
**C.** CHF 1 200
**D.** CHF 750
**E.** None of the above

**Hint.** R = p × q, with q from the demand curve.

**Answer: D.** q(15) = 50, R = 15 × 50 = **750**. 1 200 uses q = 80, ignoring the price.

### 100 The best ticket price

A cinema with 300 seats sells out at CHF 12. Each franc added to the price loses 15 spectators. Which price maximises revenue?

**A.** CHF 16
**B.** CHF 12
**C.** CHF 20
**D.** CHF 32
**E.** None of the above

**Hint.** $q = 300 - 15(p - 12) = 480 - 15p$. $R = p \cdot q$ is a parabola.

**Answer: A.** $R = 480p - 15p^2$, $R' = 480 - 30p = 0 \Leftrightarrow$ **$p = 16$** (240 seats, CHF 3 840 against 3 600 at 12). 32 is where nobody comes.

### 101 What a budget buys

Total cost is $C(q) = 5\,000 + 12q$. How many units can a budget of CHF 11 000 produce?

**A.** ≈ 1 333
**B.** 500
**C.** 600
**D.** ≈ 916.7
**E.** None of the above

**Hint.** Set $C(q) = 11\,000$.

**Answer: B.** $12q = 6\,000 \Leftrightarrow$ **$q = 500$**. 916.7 forgets that 5 000 of the budget goes to fixed costs.

### 102 Rent goes up

Price CHF 30, unit cost CHF 18, fixed costs CHF 6 000. Fixed costs rise by 20%. What is the new break-even quantity?

**A.** 600
**B.** 500
**C.** 520
**D.** 240
**E.** None of the above

**Hint.** Break-even = fixed / (price − unit cost), with the new fixed cost.

**Answer: A.** New fixed 7 200; 7 200 / 12 = **600** (from 500). A 20% rise in fixed costs moves break-even by exactly 20%.

### 103 Markup, then a discount

A shop buys an item for CHF 80 and prices it 25% above cost. In the sales it then takes 25% off that price. What is the profit per item?

**A.** CHF 20
**B.** CHF 5
**C.** −CHF 5 (a loss)
**D.** CHF 0
**E.** None of the above

**Hint.** Price: 80 × 1.25. Sale price: that × 0.75.

**Answer: C.** 80 × 1.25 = 100; 100 × 0.75 = 75; 75 − 80 = **−5**. 0 assumes +25% and −25% cancel; they are 25% of different amounts.

### 104 Compound interest

CHF 5 000 is invested at 3% a year, compounded yearly. What is it worth after 4 years?

**A.** ≈ CHF 5 450
**B.** ≈ CHF 5 627.54
**C.** CHF 5 600
**D.** ≈ CHF 5 637.48
**E.** None of the above

**Hint.** Each year multiplies by 1.03: $V = 5\,000 \times 1.03^4$.

**Answer: B.** $1.03^4 = 1.12551 \rightarrow$ **5 627.54**. 5 600 is simple interest (3% of 5 000, four times); 5 637.48 is continuous compounding.

### 105 Break-even revenue

A shop's fixed costs are CHF 18 000 a month, and each franc of sales leaves 40 centimes after variable costs. What monthly revenue breaks even?

**A.** CHF 45 000
**B.** CHF 18 000
**C.** CHF 25 200
**D.** CHF 30 000
**E.** None of the above

**Hint.** Each franc of revenue covers 0.40 of fixed cost. How many francs cover 18 000?

**Answer: A.** $0.40 \cdot R = 18\,000 \Leftrightarrow$ **R = CHF 45 000**. 7 200 multiplies instead of dividing; 30 000 divides by the 60% that goes to variable costs.

### 106 Two break-even points

A price-setter's profit is $P(q) = -q^2 + 14q - 40$. At which outputs does it break even?

**A.** q = 5 and q = 8
**B.** q = −10 and q = −4
**C.** q = 7 only
**D.** q = 2 and q = 20
**E.** None of the above

**Hint.** Break-even means P(q) = 0. Factor $-(q^2 - 14q + 40)$.

**Answer: E.** $q^2 - 14q + 40 = (q - 4)(q - 10)$: **q = 4 and q = 10**. Profit is positive between them and peaks at q = 7 (P = 9) — 7 is the best output, not a break-even point.

> On this card the right answer — **q = 4 and q = 10** — is not among A–D, so the answer is **E · None of the above**.

### 107 A curved supply

Demand is $p = 50 - q$ and supply is $p = 10 + q^2/2$ ($q \geq 0$). What is the market equilibrium?

**A.** q = 6, p = 44
**B.** q = −10, p = 60
**C.** q = 10, p = 40
**D.** q = 40, p = 10
**E.** None of the above

**Hint.** Set the two prices equal: a quadratic in q. Keep the root that makes sense.

**Answer: E.** $50 - q = 10 + q^2/2 \Leftrightarrow q^2 + 2q - 80 = 0 \Leftrightarrow (q + 10)(q - 8) = 0$. $q \geq 0$ gives **q = 8, p = 42** (check: 10 + 32 = 42). q = −10 is the root a market cannot have.

> On this card the right answer — **q = 8, p = 42** — is not among A–D, so the answer is **E · None of the above**.

### 108 A price floor

Demand is q = 100 − 2p and supply is q = 3p − 20. The authorities set a minimum price of CHF 30. How many units are left unsold?

**A.** 30
**B.** 0
**C.** 70
**D.** 52
**E.** None of the above

**Hint.** At p = 30, compute what buyers want and what sellers offer. Is the floor above the equilibrium price?

**Answer: A.** Equilibrium: 100 − 2p = 3p − 20 ⇔ p = 24, so the floor binds. At p = 30 buyers take 40 and sellers offer 70: a surplus of **30** units. 52 is the equilibrium quantity.

### 109 Two machines

Machine A costs CHF 5 000 a year plus CHF 8 per unit; machine B costs CHF 11 000 a year plus CHF 5 per unit. At what yearly volume do they cost the same?

**A.** ≈ 1 231 units
**B.** 2 000 units
**C.** 750 units
**D.** 3 000 units
**E.** None of the above

**Hint.** Write both costs as lines in q and set them equal.

**Answer: B.** 5 000 + 8q = 11 000 + 5q ⇔ 3q = 6 000 ⇔ **q = 2 000**. Above 2 000 units the higher fixed cost of B pays off. 1 231 adds the costs instead of subtracting them.

### 110 Price up, volume down

A café raises its price by 10% and sells 10% fewer coffees. What happens to its revenue?

**A.** It falls by 1%
**B.** It falls by 10%
**C.** It rises by 1%
**D.** It is unchanged
**E.** None of the above

**Hint.** Revenue = p · q. Multiply the two factors 1.10 and 0.90.

**Answer: A.** 1.10 × 0.90 = 0.99: revenue **falls by 1%**. Percentages do not simply add — +10% and −10% do not cancel.

## Derivative rules · 36 cards

### 111 A power and a reciprocal

What is the derivative of $f(x) = 5x^4 - \frac{3}{x}$ ?

**A.** $20x^3 + \frac{3}{x}$

**B.** $20x^3 - 3$

**C.** $20x^3 - \frac{3}{x^2}$

**D.** $5x^3 + \frac{3}{x^2}$

**E.** None of the above

**Hint.** Write $3/x$ as $3x^{-1}$, then use the power rule on both terms.

**Answer: E.** $(5x^4)' = 20x^3$; $(-3x^{-1})' = -3\cdot(-1)x^{-2} = +3/x^2$. **$f'(x) = 20x^3 + 3/x^2$.** Two minus signs make a plus.

> On this card the right answer — **$20x^3 + \frac{3}{x^2}$** — is not among A–D, so the answer is **E · None of the above**.

### 112 A negative power

What is the derivative of $f(x) = \frac{4}{x^3}$ ?

**A.** $-\frac{4}{3x^4}$

**B.** $-\frac{12}{x^4}$

**C.** $-\frac{12}{x^2}$

**D.** $\frac{12}{x^4}$

**E.** None of the above

**Hint.** $4/x^3 = 4x^{-3}$. The exponent goes down by one: $-3 - 1 = -4$.

**Answer: B.** $(4x^{-3})' = -12x^{-4} =$ **$-12/x^4$**. $-12/x^2$ comes from moving the exponent the wrong way.

### 113 x times its own root

What is the derivative of $f(x) = x\sqrt{x}$ ?

**A.** $\sqrt{x}$

**B.** $\frac{3}{2}x$

**C.** $\frac{3}{2}\sqrt{x}$

**D.** $\frac{1}{2\sqrt{x}}$

**E.** None of the above

**Hint.** $x\sqrt{x} = x^1 \cdot x^{1/2} = x^{3/2}$. Then the power rule.

**Answer: C.** $(x^{3/2})' = (3/2)x^{1/2} =$ **$(3/2)\sqrt{x}$**. Or by the product rule: $\sqrt{x} + x/(2\sqrt{x}) = \sqrt{x} + \sqrt{x}/2$ — the same.

### 114 A product of two polynomials

What is the derivative of $f(x) = (x^2 + 1)(x - 3)$ ?

**A.** $3x^2 + 1$

**B.** $2x$

**C.** $3x^2 - 6x - 1$

**D.** $3x^2 - 6x + 1$

**E.** None of the above

**Hint.** Product rule: $u'v + uv'$ — or multiply out first.

**Answer: D.** $2x(x - 3) + (x^2 + 1)\cdot 1 = 2x^2 - 6x + x^2 + 1 =$ **$3x^2 - 6x + 1$**. $2x$ is $(x^2+1)' \cdot (x-3)'$ — the derivative of a product is NOT the product of the derivatives.

## 115 A quotient

What is the derivative of $f(x) = \dfrac{x^2}{x - 1}$ ?

**A.** $\dfrac{3x^2 - 2x}{(x - 1)^2}$

**B.** $\dfrac{x^2 - 2x}{x - 1}$

**C.** $\dfrac{x^2 - 2x}{(x - 1)^2}$

**D.** $2x$

**E.** None of the above

**Hint.** $(u/v)' = (u'v - uv')/v^2$. The order in the numerator matters.

**Answer: C.** $(2x(x - 1) - x^2 \cdot 1)/(x - 1)^2 = (2x^2 - 2x - x^2)/(x - 1)^2 = \mathbf{(x^2 - 2x)/(x - 1)^2}$. Reversing $u'v$ and $uv'$ flips the sign.

## 116 A root of a linear function

What is the derivative of $f(x) = \sqrt{4x + 1}$ ?

**A.** $\dfrac{4}{\sqrt{4x + 1}}$

**B.** $\dfrac{1}{\sqrt{4x + 1}}$

**C.** $2\sqrt{4x + 1}$

**D.** $\dfrac{2}{\sqrt{4x + 1}}$

**E.** None of the above

**Hint.** Chain rule: $(\sqrt{u})' = u'/(2\sqrt{u})$, with $u = 4x + 1$.

**Answer: D.** $4/(2\sqrt{(4x + 1)}) = \mathbf{2/\sqrt{(4x + 1)}}$. $1/(2\sqrt{(4x+1)})$ forgets the inner derivative 4.

## 117 e to a polynomial

What is the derivative of $f(x) = e^{x^2 - 3x}$ ?

**A.** $(2x - 3)\, e^{2x - 3}$

**B.** $(x^2 - 3x)\, e^{x^2 - 3x - 1}$

**C.** $2x\, e^{x^2 - 3x}$

**D.** $(2x - 3)\, e^{x^2 - 3x}$

**E.** None of the above

**Hint.** $(e^u)' = u'\, e^u$. The exponent itself does NOT change.

**Answer: D.** $u = x^2 - 3x$, $u' = 2x - 3$: $\mathbf{(2x - 3)e^{x^2 - 3x}}$. Treating $e^u$ like a power $(u \cdot e^{u-1})$ is the classic mistake.

## 118 ln x over x

What is the derivative of $f(x) = \dfrac{\ln x}{x}$ ?

**A.** $\dfrac{1}{x} - \ln x$

**B.** $\dfrac{1}{x^2}$

**C.** $\dfrac{1 - \ln x}{x^2}$

**D.** $\dfrac{\ln x - 1}{x^2}$

**E.** None of the above

**Hint.** Quotient rule with $u = \ln x$ ($u' = 1/x$) and $v = x$ ($v' = 1$).

**Answer: C.** $((1/x) \cdot x - \ln x \cdot 1)/x^2 = \mathbf{(1 - \ln x)/x^2}$. It is zero at $x = e$, where $\ln x/x$ is largest.

## 119 A cube times a log

What is the derivative of $f(x) = x^3 \ln x$ ?

**A.** $x^2(3 \ln x - 1)$

**B.** $3x^2 \ln x + x^2$

**C.** $3x^2 \ln x$

**D.** $3x$

**E.** None of the above

**Hint.** Product rule: $(x^3)' \ln x + x^3 (\ln x)'$.

**Answer: B.** $3x^2 \ln x + x^3 \cdot (1/x) =$ **$3x^2 \ln x + x^2$**. $3x$ is $(x^3)' \times (\ln x)'$, the forbidden shortcut.

## 120 Exponential decay

What is the derivative of $f(x) = 5e^{-0.2x}$ ?

**A.** $-0.2e^{-0.2x}$

**B.** $5e^{-0.2x}$

**C.** $-e^{-0.2x}$

**D.** $-5e^{-0.2x}$

**E.** None of the above

**Hint.** $(e^{kx})' = k\, e^{kx}$. Keep the 5.

**Answer: C.** $5 \times (-0.2)\, e^{-0.2x} =$ **$-e^{-0.2x}$**. $-0.2e^{-0.2x}$ drops the factor 5.

## 121 Growth at 5% a period

What is the derivative of $f(x) = 10 \cdot 1.05^x$ ?

**A.** $10 \cdot 1.05^x \cdot \ln 1.05$

**B.** $10 \cdot 1.05^x$

**C.** $\dfrac{10 \cdot 1.05^x}{\ln 1.05}$

**D.** $10x \cdot 1.05^{x-1}$

**E.** None of the above

**Hint.** $(b^x)' = b^x \ln b$. x is in the EXPONENT, so the power rule does not apply.

**Answer: A.** **$10 \cdot 1.05^x \cdot \ln 1.05$** $\approx 0.488 \cdot 1.05^x$. $x \cdot 1.05^{x-1}$ treats the exponential like a power.

## 122 One over a square

What is the derivative of $f(x) = \dfrac{1}{(2x+5)^2}$ ?

**A.** $-\dfrac{2}{(2x+5)^3}$

**B.** $-\dfrac{4}{2x+5}$

**C.** $-\dfrac{4}{(2x+5)^3}$

**D.** $\dfrac{4}{(2x+5)^3}$

**E.** None of the above

**Hint.** Write it as $(2x+5)^{-2}$. Power rule, then multiply by the inside derivative 2.

**Answer: C.** $-2(2x+5)^{-3} \cdot 2 =$ **$-4/(2x+5)^3$**. $-2/(2x+5)^3$ forgets the inner 2.

### 123 ln of a root

What is the derivative of $f(x) = \ln(\sqrt{x})$ ?

**A.** $\frac{1}{2x}$ **B.** $\frac{2}{x}$

**C.** $\frac{1}{2\sqrt{x}}$ **D.** $\frac{1}{\sqrt{x}}$

**E.** None of the above

**Hint.** Simplify first: $\ln \sqrt{x} = \frac{1}{2} \ln x$.

**Answer: A.** $\frac{1}{2} \cdot (1/x) = \mathbf{1/(2x)}$. By the chain rule: $(1/\sqrt{x}) \cdot 1/(2\sqrt{x}) = 1/(2x)$ — the same.

### 124 A second derivative

Let $f(x) = x^4 - 2x^3 + 5x$. What is $f''(x)$?

**A.** $12x - 12$ **B.** $12x^2 - 12x$

**C.** $12x^2 - 12x + 5$ **D.** $4x^3 - 6x^2 + 5$

**E.** None of the above

**Hint.** Differentiate twice. The 5x disappears at the second step.

**Answer: B.** $f' = 4x^3 - 6x^2 + 5$, $f'' = \mathbf{12x^2 - 12x}$. The first option stops after one derivative.

### 125 A second derivative at a point

Let $f(x) = x\,e^x$. Compute $f''(0)$.

**A.** 2 **B.** e

**C.** 0 **D.** 1

**E.** None of the above

**Hint.** $f' = e^x + x\,e^x = (x + 1)e^x$. Differentiate once more.

**Answer: A.** $f'' = e^x + (x + 1)e^x = (x + 2)e^x$, so $f''(0) = \mathbf{2}$. $f'(0) = 1$ is one step short.

### 126 A derivative at a point

Let $f(x) = x^2 + \frac{16}{x}$. Compute $f'(2)$.

**A.** 8 **B.** 0

**C.** 4 **D.** −4

**E.** None of the above

**Hint.** $f'(x) = 2x - 16/x^2$. Then put $x = 2$.

**Answer: B.** $f'(2) = 4 - 16/4 = \mathbf{0}$ — the graph is flat there; $x = 2$ is where $x^2 + 16/x$ is smallest for $x > 0$.

### 127 A ratio of two lines

What is the derivative of $f(x) = \dfrac{1+x}{1-x}$ ?

**A.** $\dfrac{2}{1-x}$ **B.** $0$

**C.** $\dfrac{2x}{(1-x)^2}$ **D.** $\dfrac{2}{(1-x)^2}$

**E.** None of the above

**Hint.** Quotient rule: $u = 1+x$, $v = 1-x$, $v' = -1$.

**Answer: D.** $(1\cdot(1-x) - (1+x)(-1))/(1-x)^2 = (1-x+1+x)/(1-x)^2 = \mathbf{2/(1-x)^2}$.

### 128 x times e to the minus x

What is the derivative of $f(x) = x\,e^{-x}$ ?

**A.** $e^{-x}$ **B.** $(1+x)\,e^{-x}$

**C.** $(x-1)\,e^{-x}$ **D.** $-x\,e^{-x}$

**E.** None of the above

**Hint.** Product rule; $(e^{-x})' = -e^{-x}$.

**Answer: E.** $1\cdot e^{-x} + x\cdot(-e^{-x}) = \mathbf{(1-x)e^{-x}}$. $(1+x)e^{-x}$ forgets the minus of the chain rule.

> On this card the right answer — **$(1-x)\,e^{-x}$** — is not among A–D, so the answer is **E · None of the above**.

### 129 Divide first

What is the derivative of $f(x) = \dfrac{x^2-4}{x}$ ?

**A.** $1 + \dfrac{4}{x}$ **B.** $1 - \dfrac{4}{x^2}$

**C.** $1 + \dfrac{4}{x^2}$ **D.** $x - \dfrac{4}{x^2}$

**E.** None of the above

**Hint.** Split the fraction: $x - 4/x$. Then the power rule.

**Answer: C.** $(x - 4x^{-1})' = 1 + 4x^{-2} = \mathbf{1 + 4/x^2}$. $2x$ differentiates only the numerator.

### 130 $e^x$ times a polynomial

What is the derivative of $f(x) = e^x(x^2+1)$ ?

**A.** $2x\,e^x$ **B.** $e^x(x^2+1)$

**C.** $e^x(x+1)^2$ **D.** $2x\,e^x + 1$

**E.** None of the above

**Hint.** Product rule: $e^x(x^2+1) + e^x\cdot 2x$.

**Answer: C.** $e^x(x^2+2x+1) = \mathbf{e^x(x+1)^2}$. $e^x(x^2+1)$ forgets the derivative of the polynomial.

## 131 ln of a multiple

What is the derivative of $f(x) = \ln(5x)$ ?

**A.** $\frac{\ln 5}{x}$ **B.** $5$

**C.** $\frac{5}{x}$ **D.** $\frac{1}{x}$

**E.** None of the above

**Hint.** $\ln(5x) = \ln 5 + \ln x$, and $\ln 5$ is a constant.

**Answer: D.** $(\ln 5)' = 0$, $(\ln x)' = 1/x$: **$1/x$**. By the chain rule: $5/(5x)$ — the 5s cancel.

## 132 A cube root of a quadratic

What is the derivative of $f(x) = \sqrt[3]{(x^2 + 7)}$ ?

**A.** $\frac{2x}{3\sqrt[3]{(x^2 + 7)^2}}$ **B.** $\frac{2x}{\sqrt[3]{(x^2 + 7)^2}}$

**C.** $\frac{1}{3\sqrt[3]{(x^2 + 7)^2}}$ **D.** $\frac{2x}{3}\sqrt[3]{(x^2 + 7)}$

**E.** None of the above

**Hint.** $(x^2 + 7)^{1/3}$: power rule with exponent $1/3$, then $\times$ the inner derivative $2x$.

**Answer: A.** $(1/3)(x^2 + 7)^{-2/3} \cdot 2x = \mathbf{2x/(3\sqrt[3]{(x^2 + 7)^2})}$.

## 133 A log squared

What is the derivative of $f(x) = (\ln x)^2$ ?

**A.** $\frac{2 \ln x}{x}$ **B.** $\frac{\ln x}{x}$

**C.** $2 \ln x$ **D.** $\frac{1}{x^2}$

**E.** None of the above

**Hint.** $u^2$ with $u = \ln x$: $2u \cdot u'$.

**Answer: A.** $2 \ln x \cdot (1/x) = \mathbf{2 \ln x / x}$. $2 \ln x$ forgets the inner derivative. $(\ln x)^2$ is not $\ln(x^2) = 2 \ln x$.

## 134 An exponential over x

What is the derivative of $f(x) = \frac{e^{3x}}{x}$ ?

**A.** $3e^{3x}$ **B.** $\frac{3e^{3x}}{x}$

**C.** $\frac{e^{3x}(1 - 3x)}{x^2}$ **D.** $\frac{e^{3x}(3x - 1)}{x^2}$

**E.** None of the above

**Hint.** Quotient rule: $u = e^{3x}$ ($u' = 3e^{3x}$), $v = x$.

**Answer: D.** $(3e^{3x} \cdot x - e^{3x} \cdot 1)/x^2 = \mathbf{e^{3x}(3x - 1)/x^2}$.

### 135 A product with a square

What is the derivative of $f(x) = (x + 1)^2(x - 2)$ ?

**A.** $2(x + 1)$

**B.** $3x^2 - 3$

**C.** $(x + 1)^2$

**D.** $3x^2 + 3$

**E.** None of the above

**Hint.** Product rule, or expand first: $(x + 1)^2(x - 2) = x^3 - 3x - 2$.

**Answer: B.** Expanded: $x^3 - 3x - 2$, so $f' = \mathbf{3x^2 - 3}$. By the product rule: $2(x+1)(x-2) + (x+1)^2 = (x+1)(3x-3)$ — the same.

### 136 One over a root

What is the derivative of $f(x) = \frac{1}{\sqrt{x}}$ ?

**A.** $-\frac{1}{x\sqrt{x}}$

**B.** $\frac{1}{2\sqrt{x}}$

**C.** $-\frac{1}{2\sqrt{x}}$

**D.** $-\frac{1}{2x\sqrt{x}}$

**E.** None of the above

**Hint.** $1/\sqrt{x} = x^{-1/2}$. The new exponent is $-3/2$.

**Answer: D.** $(-1/2)x^{-3/2} = \mathbf{-1 / (2x\sqrt{x})}$. $-1/(2\sqrt{x})$ lowers the exponent to $-1/2 - 0$ instead of $-1/2 - 1$.

### 137 A third derivative

Let $f(x) = x^5$. What is $f'''(x)$?

**A.** $120x$

**B.** $20x^3$

**C.** $60x^3$

**D.** $60x^2$

**E.** None of the above

**Hint.** Three times the power rule: $5x^4$, then $20x^3$, then …

**Answer: D.** $5x^4 \to 20x^3 \to \mathbf{60x^2}$. $20x^3$ is $f''$.

### 138 Two to the 3x

What is the derivative of $f(x) = 2^{3x}$ ?

**A.** $2^{3x} \cdot \ln 2$

**B.** $3 \cdot 2^{3x}$

**C.** $3 \cdot 2^{3x} \cdot \ln 2$

**D.** $3x \cdot 2^{3x - 1}$

**E.** None of the above

**Hint.** $(b^u)' = b^u \cdot \ln b \cdot u'$, with $u = 3x$.

**Answer: C.** $2^{3x} \cdot \ln 2 \cdot 3 = \mathbf{3 \cdot 2^{3x} \cdot \ln 2}$ (= $8^x \ln 8$, the same thing). $8^x$ is the function itself, not its derivative.

## 139 $e^x$ times ln x

What is the derivative of $f(x) = e^x \ln x$ ?

**A.** $\frac{e^x}{x}$

**B.** $e^x(\ln x - \frac{1}{x})$

**C.** $e^x + \frac{1}{x}$

**D.** $e^x \ln x$

**E.** None of the above

**Hint.** Product rule: $(e^x)' \ln x + e^x (\ln x)'$.

**Answer: E.** $e^x \ln x + e^x \cdot (1/x) =$ **$e^x(\ln x + 1/x)$**. $e^x / x$ multiplies the two derivatives; $e^x + 1/x$ adds them — neither is the product rule.

> On this card the right answer — **$e^x(\ln x + \frac{1}{x})$** — is not among A–D, so the answer is **E · None of the above**.

## 140 A line over a parabola

What is the derivative of $f(x) = \frac{3x - 2}{x^2 + 1}$ ?

**A.** $\frac{-3x^2 + 4x + 3}{(x^2 + 1)^2}$

**B.** $\frac{3}{2x}$

**C.** $\frac{-3x^2 + 4x + 3}{x^2 + 1}$

**D.** $\frac{9x^2 - 4x + 3}{(x^2 + 1)^2}$

**E.** None of the above

**Hint.** $(u/v)' = (u'v - uv')/v^2$, with $u = 3x - 2$ and $v = x^2 + 1$.

**Answer: A.** $(3(x^2 + 1) - (3x - 2)\cdot 2x)/(x^2 + 1)^2 = (3x^2 + 3 - 6x^2 + 4x)/(x^2 + 1)^2 =$ **$(-3x^2 + 4x + 3)/(x^2 + 1)^2$**. $9x^2 - 4x + 3$ adds $uv'$ instead of subtracting it.

## 141 A root times an exponential

What is the derivative of $f(x) = \sqrt{x}\, e^{2x}$ ?

**A.** $e^{2x}(\frac{1}{2\sqrt{x}} + \sqrt{x})$

**B.** $e^{2x}(\frac{1}{2\sqrt{x}} + 2\sqrt{x})$

**C.** $\frac{e^{2x}}{\sqrt{x}}$

**D.** $2\sqrt{x}\, e^{2x}$

**E.** None of the above

**Hint.** Product rule, and inside it the chain rule: $(e^{2x})' = 2e^{2x}$.

**Answer: B.** $(1/(2\sqrt{x}))\cdot e^{2x} + \sqrt{x} \cdot 2e^{2x} =$ **$e^{2x}(1/(2\sqrt{x}) + 2\sqrt{x})$**. Dropping the inner 2 gives the third option; multiplying the two derivatives gives $e^{2x}/\sqrt{x}$.

## 142 The logistic curve

What is the derivative of $f(x) = \dfrac{1}{1 + e^{-x}}$ ?

**A.** $-\dfrac{e^{-x}}{(1 + e^{-x})^2}$

**B.** $-\dfrac{1}{(1 + e^{-x})^2}$

**C.** $\dfrac{e^{-x}}{1 + e^{-x}}$

**D.** $\dfrac{e^{-x}}{(1 + e^{-x})^2}$

**E.** None of the above

**Hint.** $(1/u)' = -u'/u^2$, with $u = 1 + e^{-x}$. What is $u'$?

**Answer: D.** $u' = -e^{-x}$, so $-u'/u^2 = \mathbf{e^{-x}/(1 + e^{-x})^2}$ — positive, as it must be: the curve only rises. The first option loses one of the two minus signs; $-1/u^2$ forgets $u'$ altogether.

## 143 ln of a quotient

What is the derivative of $f(x) = \ln\left(\dfrac{x + 1}{x - 1}\right)$, $x > 1$ ?

**A.** $-\dfrac{2}{x^2 - 1}$

**B.** $\dfrac{x - 1}{x + 1}$

**C.** $\dfrac{2x}{x^2 - 1}$

**D.** $\dfrac{2}{x^2 - 1}$

**E.** None of the above

**Hint.** Split it first: $\ln(x + 1) - \ln(x - 1)$.

**Answer: A.** $1/(x + 1) - 1/(x - 1) = ((x - 1) - (x + 1))/(x^2 - 1) = \mathbf{-2/(x^2 - 1)}$. $2x/(x^2 - 1)$ adds the two pieces; $-2/(x - 1)^2$ is the derivative of the quotient itself, without the ln.

## 144 A square times a power of 2

What is the derivative of $f(x) = x^2 \cdot 2^x$ ?

**A.** $2x \cdot 2^x \ln 2$

**B.** $2^x(2x + x^2)$

**C.** $2^x(2x + x^2 \ln 2)$

**D.** $2^x(2x \ln 2 + x^2)$

**E.** None of the above

**Hint.** $(a^x)' = a^x \ln a$ — not $x \cdot a^{x-1}$. Then the product rule.

**Answer: C.** $2x \cdot 2^x + x^2 \cdot 2^x \ln 2 = \mathbf{2^x(2x + x^2 \ln 2)}$. Forgetting ln 2 treats $2^x$ like $e^x$; $x^3 \cdot 2^{x-1}$ treats $2^x$ like a power of x.

## 145 e to a root

What is the derivative of $f(x) = e^{\sqrt{x}}$ ?

**A.** $\sqrt{x}\, e^{\sqrt{x}} - 1$

**B.** $e^{\sqrt{x}}$

**C.** $e^{1/(2\sqrt{x})}$

**D.** $\dfrac{e^{\sqrt{x}}}{2\sqrt{x}}$

**E.** None of the above

**Hint.** $(e^u)' = u'\, e^u$ with $u = \sqrt{x}$, $u' = 1/(2\sqrt{x})$.

**Answer: D.** $\mathbf{e^{\sqrt{x}}/(2\sqrt{x})}$. The exponent stays as it is; only the factor $u'$ appears in front. $2\sqrt{x}$ multiplies by the inverse of $u'$.

### 146 A second derivative with a log

Let $f(x) = x^2 \ln x$, $x > 0$. What is $f''(x)$?

**A.** $2x \ln x + x$

**B.** $-\frac{2}{x^2}$

**C.** $2 \ln x + 3$

**D.** $2 \ln x + 1$

**E.** None of the above

**Hint.** First $f'$ by the product rule, then differentiate $f'$ — again with the product rule on its first term.

**Answer: C.** $f' = 2x \ln x + x^2 \cdot (1/x) = 2x \ln x + x$. $f'' = 2 \ln x + 2x \cdot (1/x) + 1 = \mathbf{2 \ln x + 3}$. $2x \ln x + x$ is $f'$; $-2/x^2$ multiplies the two second derivatives.

## Tangents & marginal analysis · 18 cards

### 147 Tangent to a root

What is the tangent line to $f(x) = \sqrt{x}$ at $x = 9$?

**A.** $y = 6x - 51$

**B.** $y = \frac{x}{3}$

**C.** $y = \frac{x}{6} - \frac{3}{2}$

**D.** $y = \frac{x}{6} + 3$

**E.** None of the above

**Hint.** $y = f(9) + f'(9)(x - 9)$. $f(9) = 3$, $f'(x) = 1/(2\sqrt{x})$.

**Answer: E.** $f'(9) = 1/6$. $y = 3 + (x - 9)/6 = \mathbf{x/6 + 3/2}$. $y = x/6 + 3$ forgets to subtract $9/6$.

> On this card the right answer — $\mathbf{y = \frac{x}{6} + \frac{3}{2}}$ — is not among A–D, so the answer is **E · None of the above**.

### 148 Tangent to ln at 1

What is the tangent line to $f(x) = \ln x$ at $x = 1$?

**A.** $y = x + 1$

**B.** $y = x - 1$

**C.** $y = 1$

**D.** $y = -x + 1$

**E.** None of the above

**Hint.** $\ln 1 = 0$ and $(\ln x)' = 1/x$.

**Answer: B.** $f(1) = 0$, $f'(1) = 1$: $y = 0 + 1 \cdot (x - 1) = \mathbf{x - 1}$. Near 1, $\ln x \approx x - 1$.

### 149 Marginal cost of a cubic

$C(q) = 0.02q^3 - 3q^2 + 200q + 5\,000$. What is the marginal cost at $q = 50$?

**A.** CHF 50

**B.** CHF 350

**C.** CHF 150

**D.** CHF 7 500

**E.** None of the above

**Hint.** Marginal cost is $C'(q)$. Differentiate term by term, then put $q = 50$.

**Answer: A.** $C' = 0.06q^2 - 6q + 200$; $C'(50) = 150 - 300 + 200 = \mathbf{50}$. 150 stops after the first term.

## 150 Marginal revenue

$R(q) = 120q - 0.4q^2$. What is the marginal revenue at $q = 100$?

**A.** CHF 8 000
**B.** CHF 120
**C.** –CHF 40
**D.** CHF 80
**E.** None of the above

**Hint.** $MR = R'(q) = 120 - 0.8q$.

**Answer: E.** $R'(100) = 120 - 80 =$ **40**: the 101st unit adds about CHF 40. 8 000 is total revenue.

> On this card the right answer — **CHF 40** — is not among A–D, so the answer is **E · None of the above**.

## 151 Marginal revenue meets marginal cost

$R(q) = 60q - 0.5q^2$, $C(q) = 200 + 12q + 0.25q^2$. At which $q$ is $MR = MC$?

**A.** $q = 16$
**B.** $q = 48$
**C.** $q = 64$
**D.** $q = 32$
**E.** None of the above

**Hint.** $MR = 60 - q$, $MC = 12 + 0.5q$. Set them equal.

**Answer: D.** $60 - q = 12 + 0.5q \Leftrightarrow 1.5q = 48 \Leftrightarrow$ **$q = 32$**. 48 is 60 – 12, before dividing by 1.5.

## 152 Cheapest per unit

$C(q) = 0.5q^2 + 30q + 800$. At which output is the AVERAGE cost per unit lowest?

**A.** $q = 20$
**B.** $q = 1\,600$
**C.** $q = 80$
**D.** $q = 40$
**E.** None of the above

**Hint.** Average cost $\bar{C} = C/q = 0.5q + 30 + 800/q$. Set its derivative to zero.

**Answer: D.** $\bar{C}' = 0.5 - 800/q^2 = 0 \Leftrightarrow q^2 = 1\,600 \Leftrightarrow$ **$q = 40$** ($\bar{C} = 70$ there). At that point marginal cost equals average cost: $C'(40) = 70$.

## 153 The cost of one more unit

$C(q) = 1\,000 + 20q + 0.1q^2$. Using the marginal cost, estimate the cost of producing the 101st unit.

**A.** CHF 3 000
**B.** ≈ CHF 40
**C.** CHF 40.10
**D.** CHF 20
**E.** None of the above

**Hint.** The cost of one more unit ≈ $C'(100)$. The question says ESTIMATE using marginal cost.

**Answer: B.** $C'(q) = 20 + 0.2q$, $C'(100) =$ **40**. The exact extra cost is $C(101) - C(100) = 40.10$; the marginal cost is its approximation.

### 154 Slope of a tangent

What is the slope of the tangent to $f(x) = 3x^2 - x$ at $x = -1$?

**A.** 7
**B.** −7
**C.** −5
**D.** 4
**E.** None of the above

**Hint.** The slope of the tangent is $f'(-1)$.

**Answer: B.** $f'(x) = 6x - 1$, $f'(-1) = -6 - 1 = \mathbf{-7}$. 4 is $f(-1)$, the height, not the slope.

### 155 Tangent to a cubic

What is the tangent line to $f(x) = x^3 - 2x$ at $x = 2$?

**A.** $y = 10x + 4$
**B.** $y = 10x - 4$
**C.** $y = 10x - 16$
**D.** $y = 4x + 2$
**E.** None of the above

**Hint.** $f(2)$ and $f'(2)$, then $y = f(2) + f'(2)(x - 2)$.

**Answer: C.** $f(2) = 4$, $f' = 3x^2 - 2 \rightarrow f'(2) = 10$. $y = 4 + 10(x - 2) = \mathbf{10x - 16}$. $12x - 20$ forgets the $-2$ in $f'$.

### 156 Marginal profit

$P(q) = -0.02q^2 + 60q - 8\,000$. What is the marginal profit at $q = 1\,000$?

**A.** CHF 20
**B.** −CHF 20
**C.** CHF 40
**D.** CHF 32 000
**E.** None of the above

**Hint.** $P'(q) = -0.04q + 60$.

**Answer: A.** $P'(1\,000) = -40 + 60 = \mathbf{20}$: one more unit still adds about CHF 20. 32 000 is the profit itself.

### 157 Where marginal cost is lowest

$C(q) = q^3 - 12q^2 + 60q + 100$. At which output is the marginal cost lowest?

**A.** $q = 6$
**B.** $q = 2$
**C.** $q = 12$
**D.** $q = 4$
**E.** None of the above

**Hint.** $MC = C'(q) = 3q^2 - 24q + 60$ is a parabola. Its minimum is where its own derivative is 0.

**Answer: D.** $MC' = C'' = 6q - 24 = 0 \Leftrightarrow \mathbf{q = 4}$ ($MC = 12$ there). This is the inflection point of $C$.

### 158 How fast average cost falls

$C(q) = 2\,000 + 5q$. The average cost is $A(q) = C(q)/q$. What is $A'(100)$?

**A.** –0.2
**B.** 0.2
**C.** 5
**D.** –20
**E.** None of the above

**Hint.** $A(q) = 2\,000/q + 5$, so $A'(q) = -2\,000/q^2$.

**Answer: A.** $A'(100) = -2\,000 / 10\,000 =$ **–0.2**: the 101st unit lowers the average by about 20 centimes. 25 is A(100) itself.

### 159 Marginal revenue from a price curve

Demand $p = 200 - 4q$. What is the marginal revenue at $q = 10$?

**A.** CHF 80
**B.** CHF 1 600
**C.** CHF 160
**D.** CHF 120
**E.** None of the above

**Hint.** $R = p \cdot q = 200q - 4q^2$, then $R'$.

**Answer: D.** $R' = 200 - 8q$, $R'(10) =$ **120**. 160 is the price at $q = 10$ — marginal revenue is below price because the price falls on every unit.

### 160 Where the tangent is flat

At which point of the graph of $f(x) = x^2 - 6x + 11$ is the tangent horizontal?

**A.** (–3, 38)
**B.** (3, –2)
**C.** (2, 3)
**D.** (6, 11)
**E.** None of the above

**Hint.** Horizontal tangent: $f'(x) = 0$. Then compute f there.

**Answer: E.** $f' = 2x - 6 = 0 \Leftrightarrow x = 3$; $f(3) = 9 - 18 + 11 = 2$: **(3, 2)**.

> On this card the right answer — **(3, 2)** — is not among A–D, so the answer is **E · None of the above**.

### 161 Tangent to $e^{2x}$ at 0

What is the tangent line to $f(x) = e^{2x}$ at $x = 0$?

**A.** $y = 2x$
**B.** $y = x + 1$
**C.** $y = 2x + 2$
**D.** $y = 2x + 1$
**E.** None of the above

**Hint.** $y = f(0) + f'(0)(x - 0)$. $f'(x) = 2e^{2x}$.

**Answer: D.** $f(0) = 1$ and $f'(0) = 2$: **$y = 2x + 1$**. $y = x + 1$ is the tangent to $e^x$, not $e^{2x}$; $y = 2x$ forgets that the curve passes through (0, 1).

### 162 Tangent to x ln x at e

What is the tangent line to $f(x) = x \ln x$ at $x = e$?

**A.** $y = ex - e$
**B.** $y = 2x - e$
**C.** $y = 2x - 2e$
**D.** $y = 2x + e$
**E.** None of the above

**Hint.** $f(e) = e \cdot \ln e = e$. $f'(x) = \ln x + 1$.

**Answer: B.** $f'(e) = 2$, so $y = e + 2(x - e) =$ **$2x - e$**. $y = x$ uses $f' = \ln x$, forgetting the product rule.

### 163 The exact cost of one more unit

Total cost is $C(q) = 0.02q^2 + 5q + 300$. What does the 51st unit cost exactly, i.e. $C(51) - C(50)$?

**A.** CHF 7.04
**B.** CHF 5.02
**C.** CHF 607.02
**D.** CHF 7
**E.** None of the above

**Hint.** Compute both totals — or note that only the $q^2$ term differs from the marginal-cost estimate.

**Answer: E.** $C(51) = 52.02 + 255 + 300 = 607.02$ and $C(50) = 50 + 250 + 300 = 600$, so the 51st unit costs **CHF 7.02**. The marginal cost $C'(50) = 0.04 \times 50 + 5 = 7$ is the approximation of it; 607.02 is the total, not the extra.

> On this card the right answer — **CHF 7.02** — is not among A–D, so the answer is **E · None of the above**.

### 164 A tangent parallel to a line

At which $x$ is the tangent to $f(x) = x^2 - 4x + 1$ parallel to the line $y = 2x + 7$?

**A.** $x = 4$
**B.** $x = 1$
**C.** $x = 2$
**D.** $x = 3$
**E.** None of the above

**Hint.** Parallel lines have equal slopes. Solve $f'(x) = 2$.

**Answer: D.** $f'(x) = 2x - 4 = 2 \Leftrightarrow$ **$x = 3$**, at the point $(3, -2)$. $x = 2$ is where the tangent is flat; $x = 1$ is where the slope is $-2$.

## Chain rule in business · 11 cards

### 165 Ads → visitors → sales

Spending $a$ (thousand CHF) on ads brings $N(a) = 200\sqrt{a}$ visitors; each visitor buys on average CHF 0.05. At $a = 25$, how fast do sales grow per extra thousand spent?

**A.** CHF 50
**B.** CHF 2
**C.** CHF 0.5
**D.** CHF 1 per thousand
**E.** None of the above

**Hint.** Sales $S = 0.05 \cdot N(a) = 10\sqrt{a}$. Differentiate with respect to $a$.

**Answer: D.** $dS/da = 0.05 \cdot N'(a) = 0.05 \cdot 200 / (2\sqrt{a}) = 5/\sqrt{a}$; at $a = 25$: $5/5 =$ **1**. 50 is the level of sales, not the rate.

### 166 A price that drifts

The price rises over time as $p(t) = 20 + 0.5t$ ($t$ in weeks) and demand is $q(p) = 1\,000 - 30p$. How fast does demand change, per week?

**A.** −15 units per week
**B.** 0.5
**C.** +15
**D.** −600
**E.** None of the above

**Hint.** $dq/dt = dq/dp \times dp/dt$.

**Answer: A.** $dq/dp = -30$, $dp/dt = 0.5$: $dq/dt =$ **−15** units per week. −30 is the change per franc, not per week.

### 167 Cost through time

Output grows as $q(t) = 50 + 4t$ ($t$ in months), and cost is $C(q) = 1\,000 + 2q^2$. How fast is cost rising at $t = 5$?

**A.** CHF 1 400
**B.** CHF 1 120 per month
**C.** CHF 9 800
**D.** CHF 280
**E.** None of the above

**Hint.** $dC/dt = C'(q) \cdot q'(t)$. First find $q$ at $t = 5$.

**Answer: B.** $q(5) = 70$, $C'(q) = 4q = 280$, $q' = 4$. $dC/dt = 280 \times 4 =$ **1 120** a month. 280 is per unit of output, not per month.

### 168 A cubed demand curve

Demand over time is $D(t) = (2t + 1)^3$. Compute $D'(1)$.

**A.** 9
**B.** 27
**C.** 54
**D.** 162
**E.** None of the above

**Hint.** Chain rule: $3(2t + 1)^2 \times 2$.

**Answer: C.** $D'(t) = 6(2t + 1)^2$; $D'(1) = 6 \times 9 =$ **54**. 27 is $D(1)$; 27 also comes from $3 \times 9$ without the inner 2.

### 169 Costs in euros, paid in francs

A supplier's cost in euros is $E(q) = 200 + 3q$, and 1 euro costs CHF 0.95. What is the marginal cost in francs?

**A.** CHF 190
**B.** CHF 0.95
**C.** CHF 2.85 per unit
**D.** CHF 3.95
**E.** None of the above

**Hint.** $C(q) = 0.95 \cdot E(q)$. $dC/dq = 0.95 \cdot E'(q)$.

**Answer: C.** $0.95 \times 3 =$ **2.85** francs per unit. 3 is the marginal cost in euros.

### 170 Warmer days, more ice cream

Ice-cream sales are $S(T) = 40T$ ($T$ in °C), and the temperature rises as $T(t) = 18 + 2t$ ($t$ in hours after 8 am). How fast do sales grow per hour?

**A.** 80 per hour
**B.** 720
**C.** 2
**D.** 40
**E.** None of the above

**Hint.** $dS/dt = dS/dT \times dT/dt$.

**Answer: A.** $40 \times 2 =$ **80** per hour. 40 is per degree, not per hour.

### 171 A root of a sum of squares

$R(q) = \sqrt{q^2 + 900}$. Compute $R'(40)$.

**A.** 0.5
**B.** 50
**C.** 40
**D.** 0.8
**E.** None of the above

**Hint.** $(\sqrt{u})' = u' / (2\sqrt{u})$, with $u = q^2 + 900$.

**Answer: D.** $R' = 2q / (2\sqrt{(q^2 + 900)}) = q / \sqrt{(q^2 + 900)}$; at 40: $40 / 50 =$ **0.8**. 1.6 forgets the 2 in $2\sqrt{u}$.

### 172 Hiring over time

Output is $q(L) = 10\sqrt{L}$ with $L$ workers, and the workforce grows as $L(t) = 100 + 4t$ ($t$ in months). How fast does output grow at $t = 0$?

**A.** 4
**B.** 2 units per month
**C.** 0.2
**D.** 20
**E.** None of the above

**Hint.** $dq/dt = q'(L) \cdot L'(t)$, with $q'(L) = 5/\sqrt{L}$ and $L(0) = 100$.

**Answer: B.** $q'(100) = 5/10 = 0.5$ per worker; $L' = 4$ workers a month $\rightarrow$ **2** units a month.

### 173 Cost through the working day

Cost is $C(q) = 400 + 2q^2$, and output after $t$ hours is $q(t) = 5t + 10$. How fast is cost rising at $t = 2$ (CHF per hour)?

**A.** 5
**B.** 1 200
**C.** 400
**D.** 100
**E.** None of the above

**Hint.** $dC/dt = C'(q) \cdot q'(t)$. First find $q$ at $t = 2$.

**Answer: C.** $q(2) = 20$, $C'(q) = 4q = 80$ per unit, $q' = 5$ units an hour: $80 \times 5 =$ **CHF 400 an hour**. 80 stops at $C'(q)$; 1 200 is $C(20)$ itself.

### 174 A price that rises week by week

Revenue as a function of price is $R(p) = p(100 - 2p)$, and the price rises as $p(t) = 10 + 0.5t$ (t in weeks). How fast does revenue change at $t = 10$?

**A.** 20 CHF per week
**B.** 1 050
**C.** –20
**D.** 30
**E.** None of the above

**Hint.** $dR/dt = R'(p) \cdot p'(t)$, with $R'(p) = 100 - 4p$. Use the price at $t = 10$, not at $t = 0$.

**Answer: A.** $p(10) = 15$, $R'(15) = 100 - 60 = 40$, $p' = 0.5$: $dR/dt =$ **20** francs a week. 30 uses $R'$ at $p = 10$ (the price at $t = 0$); 1 050 is $R(15)$.

### 175 Users and revenue

An app has $U(t) = 1\,000 \ln(1 + t)$ users after t months, and each user brings CHF 3 a month. How fast is monthly revenue growing at $t = 4$?

**A.** CHF 600 per month
**B.** CHF 200
**C.** CHF 3 000
**D.** CHF 750
**E.** None of the above

**Hint.** Revenue $R = 3U$. $dR/dt = 3 \cdot U'(t)$, with $U'(t) = 1\,000/(1 + t)$.

**Answer: A.** $U'(4) = 1\,000/5 = 200$ users a month; × CHF 3 = **CHF 600** a month. 200 forgets the price per user; 750 divides by t instead of $1 + t$; 4 828 is the revenue level, not its growth.

## Approximation & Newton · 15 cards

### 176 A root without a calculator

Use the linear approximation of $f(x) = \sqrt{x}$ at $x_0 = 100$ to estimate $\sqrt{102}$.

**A.** 10.01
**B.** 10.1
**C.** 10.2
**D.** 10.0995
**E.** None of the above

**Hint.** $f(x) \approx f(100) + f'(100)(x - 100)$, with $f'(x) = 1/(2\sqrt{x})$.

**Answer: B.** $f'(100) = 1/20 = 0.05$. $\sqrt{102} \approx 10 + 0.05 \times 2 =$ **10.1**. 10.0995 is the true value — close, but not what the approximation gives.

### 177 Linear approximation of 1/x

What is the linear approximation of $f(x) = \frac{1}{x}$ at $x_0 = 2$?

**A.** $y = \frac{x}{4}$

**B.** $y = -\frac{x}{2} + 1$

**C.** $y = \frac{x}{4} + 1$

**D.** $y = -\frac{x}{4} + \frac{1}{2}$

**E.** None of the above

**Hint.** $f(2) = 1/2$, $f'(x) = -1/x^2$, so $f'(2) = -1/4$.

**Answer: E.** $y = \frac{1}{2} - \frac{1}{4}(x - 2) = \frac{1}{2} - \frac{x}{4} + \frac{1}{2} = \mathbf{-\frac{x}{4} + 1}$. $-x/4 + 1/2$ forgets the $+ 1/2$ that comes from $-1/4 \times (-2)$.

> On this card the right answer — $\mathbf{y = -\frac{x}{4} + 1}$ — is not among A–D, so the answer is **E · None of the above**.

### 178 Estimate a cost a little further out

At $q = 200$ a firm's cost is CHF 9 000 and its marginal cost is CHF 35. Estimate the cost at $q = 204$.

**A.** CHF 9 004

**B.** CHF 9 035

**C.** CHF 9 140

**D.** CHF 9 175

**E.** None of the above

**Hint.** $C(204) \approx C(200) + C'(200) \times 4$.

**Answer: C.** $9\,000 + 35 \times 4 = \mathbf{9\,140}$. 9 035 adds one unit's cost instead of four.

### 179 Newton's method on a square root

Apply one step of Newton's method to $f(x) = x^2 - 5$, starting from $x_0 = 2$. What is $x_1$?

**A.** 2.2

**B.** 2.25

**C.** 2.236

**D.** 1.75

**E.** None of the above

**Hint.** $x_1 = x_0 - f(x_0)/f'(x_0)$, with $f'(x) = 2x$.

**Answer: B.** $f(2) = -1$, $f'(2) = 4$. $x_1 = 2 - (-1)/4 = \mathbf{2.25}$. The true root $\sqrt{5} \approx 2.236$ is the target, not the first step. 1.75 is a sign slip.

### 180 Newton on an exponential

One step of Newton's method on $f(x) = e^x - 3$, starting from $x_0 = 1$. What is $x_1$ (to 4 decimals)?

**A.** 1.5000

**B.** 1.2817

**C.** 1.0986

**D.** 0.9000

**E.** None of the above

**Hint.** $f(1) = e - 3$, $f'(x) = e^x$, so $f'(1) = e$.

**Answer: E.** $x_1 = 1 - (e - 3)/e = 3/e = \mathbf{1.1036}$. 1.0986 = ln 3 is the exact root; one step lands 0.005 away.

> On this card the right answer — **1.1036** — is not among A–D, so the answer is **E · None of the above**.

### 181 Two Newton steps

Newton's method on $f(x) = x^3 - 10$ from $x_0 = 2$. What is $x_2$ (to 4 decimals)?

**A.** 2.1000
**B.** 2.2000
**C.** 2.1545
**D.** 2.1667
**E.** None of the above

**Hint.** Two steps of $x - f(x)/f'(x)$, with $f'(x) = 3x^2$. Keep all decimals between the steps.

**Answer: C.** $x_1 = 2 - (-2)/12 = 2.16667$. $f(x_1) = 0.17130$, $f'(x_1) = 14.0833$; $x_2 = 2.16667 - 0.01216 =$ **2.1545**. 2.1667 is $x_1$. The true root $\sqrt[3]{10} = 2.15443$ — two steps already agree to four figures.

### 182 A revenue change, approximated

$R(q) = 50q - 0.05q^2$. Output rises from 400 to 408. Using $R'(400)$, estimate the change in revenue.

**A.** CHF 160
**B.** CHF 76.80
**C.** ≈ CHF 80
**D.** CHF 8
**E.** None of the above

**Hint.** $\Delta R \approx R'(400) \times \Delta q$.

**Answer: C.** $R'(q) = 50 - 0.1q$, $R'(400) = 10$; $\Delta R \approx 10 \times 8 =$ **80**. The exact change, 76.80, is what the approximation estimates.

### 183 A power close to 1

Use the linear approximation of $f(x) = x^{10}$ at $x_0 = 1$ to estimate $1.02^{10}$.

**A.** 1.2
**B.** 1.02
**C.** 1.0
**D.** 1.25
**E.** None of the above

**Hint.** $f(1) = 1$, $f'(x) = 10x^9$, so $f'(1) = 10$.

**Answer: A.** $1.02^{10} \approx 1 + 10 \times 0.02 =$ **1.2** (the true value is 1.219 — the approximation errs on the low side because $x^{10}$ curves up).

### 184 ln of a number near 1

Using the tangent to $\ln x$ at $x_0 = 1$, estimate $\ln 1.05$.

**A.** 0.05
**B.** 0.0488
**C.** 0.005
**D.** 1.05
**E.** None of the above

**Hint.** The tangent at 1 is $y = x - 1$.

**Answer: A.** $\ln 1.05 \approx 1.05 - 1 =$ **0.05**. 0.0488 is the exact value — the approximation, not the calculator, is asked for.

### 185 Newton on a quadratic

One step of Newton's method on $f(x) = x^2 - 2x - 5$, starting from $x_0 = 4$. What is $x_1$?

**A.** 3.25
**B.** 3.449
**C.** 3
**D.** 4.5
**E.** None of the above

**Hint.** $f(4) = 3$, $f'(x) = 2x - 2$, so $f'(4) = 6$.

**Answer: E.** $x_1 = 4 - 3/6 = \mathbf{3.5}$. 4.5 adds the correction instead of subtracting it; $3.449 = 1 + \sqrt{6}$ is the root itself.

> On this card the right answer — **3.5** — is not among A–D, so the answer is **E · None of the above**.

### 186 When Newton cannot start

Newton's method is applied to $f(x) = x^2 - 4$ from $x_0 = 0$. What happens at the first step?

**A.** It converges slowly to 2
**B.** $x_1 = 2$ straight away
**C.** $x_1 = 4$
**D.** The step is undefined, because $f'(0) = 0$
**E.** None of the above

**Hint.** $x_1 = x_0 - f(x_0)/f'(x_0)$. What is $f'(0)$?

**Answer: D.** $f'(x) = 2x$, $f'(0) = 0$: the formula divides by zero — **the step is undefined**. Geometrically, the tangent at $x = 0$ is horizontal and never meets the x-axis. Start from any $x_0 \neq 0$.

### 187 A square root near 100

Using the tangent to $\sqrt{x}$ at $x_0 = 100$, estimate $\sqrt{102}$.

**A.** 10.1
**B.** 11
**C.** 10.01
**D.** 10.0995
**E.** None of the above

**Hint.** $f(100) = 10$ and $f'(x) = 1/(2\sqrt{x})$, so $f'(100) = 1/20$.

**Answer: A.** $\sqrt{102} \approx 10 + 2 \times (1/20) = \mathbf{10.1}$. 10.0995 is the calculator's value — the question asks for the tangent's estimate; 10.2 forgets the 2 in $2\sqrt{x}$.

### 188 ln 3 from the tangent at e

Using the tangent to $\ln x$ at $x_0 = e$, estimate $\ln 3$.

**A.** $3/e \approx 1.104$
**B.** $1 + 3/e \approx 2.104$
**C.** $3 - e \approx 0.282$
**D.** 2
**E.** None of the above

**Hint.** $\ln e = 1$ and $(\ln x)' = 1/x$, so the slope at e is $1/e$.

**Answer: A.** $\ln x \approx 1 + (x - e)/e = x/e$, so $\ln 3 \approx \mathbf{3/e \approx 1.104}$ — close to the true 1.099 because 3 is close to $e \approx 2.718$. $3 - e$ forgets the value $\ln e = 1$.

### 189 Newton towards √10

One step of Newton's method on $f(x) = x^2 - 10$, starting from $x_0 = 3$. What is $x_1$?

**A.** $\sqrt{10} \approx 3.1623$
**B.** $10/3$
**C.** $19/6 \approx 3.1667$
**D.** 3.5
**E.** None of the above

**Hint.** $x_1 = x_0 - f(x_0)/f'(x_0)$. $f(3) = -1$ and $f'(3) = 6$.

**Answer: C.** $x_1 = 3 - (-1)/6 =$ **$19/6 \approx 3.1667$**, already within 0.005 of $\sqrt{10}$. $17/6$ loses the minus sign of $f(3)$.

### 190 Newton on a cubic

One step of Newton's method on $f(x) = x^3 - x - 1$, starting from $x_0 = 1$. What is $x_1$?

**A.** 2
**B.** 1.25
**C.** 0.5
**D.** 1.3247
**E.** None of the above

**Hint.** $f(1) = -1$, $f'(x) = 3x^2 - 1$, so $f'(1) = 2$.

**Answer: E.** $x_1 = 1 - (-1)/2 =$ **1.5**. 0.5 adds the wrong sign; 1.3247 is the root itself, which Newton only reaches after several steps.

> On this card the right answer — **1.5** — is not among A–D, so the answer is **E · None of the above**.

## Elasticity · 12 cards

### 191 Elasticity of a linear demand

Demand $q = 200 - 4p$. What is the price elasticity of demand at $p = 30$?

**A.** −4
**B.** −1.5
**C.** −0.6
**D.** 1.5
**E.** None of the above

**Hint.** $\eta = q'(p) \times p/q$. First $q(30)$.

**Answer: B.** $q(30) = 80$, $q' = -4$. $\eta = -4 \times 30/80 =$ **−1.5**: elastic. −0.6 inverts $p/q$.

### 192 A constant elasticity

Demand $q = \dfrac{1\,200}{p}$. What is the price elasticity at $p = 8$?

**A.** 1
**B.** −8
**C.** −150
**D.** −0.125
**E.** None of the above

**Hint.** $q'(p) = -1\,200/p^2$. Then $\eta = q' \times p/q$.

**Answer: E.** $q(8) = 150$, $q'(8) = -18.75$. $\eta = -18.75 \times 8/150 =$ **−1**. For $q = k/p$ it is −1 at EVERY price: revenue $p \cdot q = 1\,200$ never changes.

> On this card the right answer — **−1** — is not among A–D, so the answer is **E · None of the above**.

### 193 Inelastic: what a price rise does

At the current price, demand for bus passes has elasticity $\eta = -0.4$. What happens to revenue if the price rises slightly?

**A.** Quantity sold rises
**B.** Revenue falls
**C.** Revenue stays the same
**D.** Revenue rises
**E.** None of the above

**Hint.** $|\eta| < 1$: quantity falls proportionally LESS than price rises.

**Answer: D.** A 1% price rise loses only about 0.4% of passes, so revenue $p \cdot q$ goes up by roughly 0.6%: **revenue rises**. Quantity still falls.

### 194 The price where demand is unit-elastic

Demand $q = 90 - 3p$. At which price is the elasticity exactly $-1$?

**A.** $p = 7.5$
**B.** $p = 10$
**C.** $p = 30$
**D.** $p = 45$
**E.** None of the above

**Hint.** $\eta = -3p/(90 - 3p)$. Set it equal to $-1$ and solve.

**Answer: E.** $-3p = -(90 - 3p) \Leftrightarrow 6p = 90 \Leftrightarrow$ **$p = 15$** — halfway to the choke price 30, which is also where revenue peaks.

> On this card the right answer — **$p = 15$** — is not among A–D, so the answer is **E · None of the above**.

### 195 Elasticity from two observations

When a gym cut its monthly fee from CHF 20 to CHF 19, membership rose from 500 to 530. Estimate the price elasticity (relative to the starting point).

**A.** $-30$
**B.** $-0.83$
**C.** $-0.06$
**D.** $-1.2$
**E.** None of the above

**Hint.** $\eta \approx (\Delta q/q)/(\Delta p/p)$: percentage change of quantity over percentage change of price.

**Answer: D.** $\Delta q/q = 30/500 = +6\%$; $\Delta p/p = -1/20 = -5\%$. $\eta \approx 6\%/(-5\%) =$ **$-1.2$**. $-0.83$ divides the other way round.

### 196 Elasticity of a curved demand

Demand $q = 500 - 2p^2$. What is the price elasticity at $p = 10$?

**A.** $\approx 1.33$
**B.** $\approx -0.13$
**C.** $-0.75$
**D.** $-40$
**E.** None of the above

**Hint.** $\eta = q'(p) \cdot p/q$, with $q' = -4p$.

**Answer: E.** $q(10) = 300$, $q'(10) = -40$. $\eta = -40 \times 10/300 =$ **$-4/3 \approx -1.33$**: elastic.

> On this card the right answer — **$\approx -1.33$** — is not among A–D, so the answer is **E · None of the above**.

## 197 Elastic demand and a price cut

At the current price, $\eta = -2.5$. What does a small price cut do to revenue?

**A.** Revenue rises
**B.** Quantity falls
**C.** Revenue stays the same
**D.** Revenue falls
**E.** None of the above

**Hint.** $|\eta| > 1$: quantity reacts more than proportionally to price.

**Answer: A.** A 1% cut brings about 2.5% more sales, so revenue rises by roughly 1.5%: **revenue rises**. With elastic demand, price and revenue move in opposite directions.

## 198 A constant-elasticity demand

Demand $q = 1\,000\,p^{-0.6}$. What is the price elasticity?

**A.** 0.6
**B.** −0.4
**C.** It depends on p
**D.** −0.6, at every price
**E.** None of the above

**Hint.** $q' = -600\,p^{-1.6}$. Compute $q' \cdot p/q$.

**Answer: D.** $q' \cdot p/q = -600\,p^{-1.6} \cdot p/(1\,000\,p^{-0.6}) =$ **−0.6**, whatever p is. For $q = k \cdot p^a$ the elasticity is always a.

## 199 Elasticity of a straight demand

Demand $q = 60 - p$. What is the price elasticity at $p = 20$?

**A.** −20
**B.** −0.5
**C.** −2
**D.** −1
**E.** None of the above

**Hint.** $\eta = q' \cdot p/q$ with $q' = -1$ and $q(20) = 40$.

**Answer: B.** $\eta = -1 \times 20/40 =$ **−0.5**: inelastic here. −2 inverts $p/q$.

## 200 Elasticity of an exponential demand

Demand is $q = 400e^{-0.02p}$. What is the price elasticity at $p = 75$?

**A.** −1.5
**B.** −0.02
**C.** −75
**D.** −0.75
**E.** None of the above

**Hint.** $\eta = q'(p) \cdot p/q$. Here $q' = -0.02\,q$, so most of the expression cancels.

**Answer: A.** $\eta = (-0.02q) \cdot p/q = -0.02p$; at $p = 75$: **−1.5** — elastic. −0.02 is the slope factor, not the elasticity.

### 201 Elasticity when price is given as a function of quantity

The demand curve is written p = 80 − 2q. What is the price elasticity of demand at q = 10?

**A.** $-\frac{1}{3}$

**B.** −0.5

**C.** 3

**D.** −12

**E.** None of the above

**Hint.** Elasticity uses dq⁄dp. Rewrite as q = 40 − p⁄2, and find the price at q = 10.

**Answer: E.** q = 40 − p⁄2, so dq⁄dp = −½; at q = 10, p = 60. η = −½ × 60⁄10 = **−3**. −12 uses dp⁄dq = −2 in place of dq⁄dp; −1⁄3 inverts p⁄q.

> On this card the right answer — **−3** — is not among A–D, so the answer is **E · None of the above**.

### 202 The price cut a target needs

At the current price, η = −1.5. Approximately what price change raises sales by 6%?

**A.** A 1.5% price cut

**B.** A 9% price cut

**C.** A 4% price rise

**D.** A 6% price cut

**E.** None of the above

**Hint.** Δq⁄q ≈ η · Δp⁄p. Solve for Δp⁄p.

**Answer: E.** 6% = −1.5 × Δp⁄p ⇔ Δp⁄p = −4%: **a 4% price cut**. −9% multiplies by η instead of dividing.

> On this card the right answer — **A 4% price cut** — is not among A–D, so the answer is **E · None of the above**.

## Critical points & concavity · 13 cards

### 203 Where the slope is zero

Let f(x) = 2x³ + 3x² − 36x. What are the critical points?

**A.** x = 3 and x = −2

**B.** x = −3 and x = 2

**C.** x = −1 and x = 6

**D.** x = 2 only

**E.** None of the above

**Hint.** Solve f′(x) = 0. Divide the quadratic by 6 first.

**Answer: B.** f′ = 6x² + 6x − 36 = 6(x² + x − 6) = 6(x + 3)(x − 2). **x = −3 and x = 2.**

### 204 Maximum or minimum?

f(x) = x³ − 6x² + 9x + 2 has critical points at x = 1 and x = 3. What is f at x = 1?

**A.** A local maximum

**B.** A local minimum

**C.** A global maximum

**D.** An inflection point

**E.** None of the above

**Hint.** Second-derivative test: the sign of f″(1).

**Answer: A.** f″ = 6x − 12, f″(1) = −6 < 0 → concave down → **local maximum** (f(1) = 6). Not global: f → +∞ as x → +∞.

### 205 Where the curvature turns

At which x does $f(x) = x^3 - 6x^2 + 5$ have an inflection point?

**A.** $x = -2$ **B.** $x = 4$

**C.** $x = 0$ **D.** $x = 2$

**E.** None of the above

**Hint.** An inflection point is where $f''$ changes sign.

**Answer: D.** $f'' = 6x - 12 = 0 \Leftrightarrow$ **$x = 2$**, and $f''$ changes from $-$ to $+$ there. $x = 0$ and $x = 4$ are the critical points ($f' = 0$) — a different thing.

### 206 Where a quartic is concave down

On which interval is $f(x) = x^4 - 6x^2$ concave down?

**A.** $]-\sqrt{6}, \sqrt{6}[$ **B.** $]-\sqrt{3}, \sqrt{3}[$

**C.** $]0, +\infty[$ **D.** $]-\infty, -1[ \cup ]1, +\infty[$

**E.** None of the above

**Hint.** Concave down where $f'' < 0$.

**Answer: E.** $f'' = 12x^2 - 12 = 12(x^2 - 1) < 0 \Leftrightarrow$ **$-1 < x < 1$**. $\pm\sqrt{3}$ are where $f' = 4x^3 - 12x$ is zero — the critical points, which say nothing about curvature.

> On this card the right answer — **$]-1, 1[$** — is not among A–D, so the answer is **E · None of the above**.

### 207 The profit-maximising output

$P(q) = -q^3 + 15q^2 - 48q - 20$. Which output maximises profit?

**A.** $q = 2$ **B.** $q = 10$

**C.** $q = 16$ **D.** $q = 8$

**E.** None of the above

**Hint.** Solve $P'(q) = 0$, then use $P''$ to tell the maximum from the minimum.

**Answer: D.** $P' = -3q^2 + 30q - 48 = -3(q^2 - 10q + 16) = -3(q - 2)(q - 8)$. $P'' = -6q + 30$: $P''(2) = 18 > 0$ (a minimum, a loss of 64), $P''(8) = -18 < 0 \rightarrow$ **$q = 8$**, $P = 44$.

### 208 The lowest value of a parabola

What is the minimum VALUE of $f(x) = x^2 - 8x + 30$?

**A.** $-14$ **B.** 14

**C.** 16 **D.** 4

**E.** None of the above

**Hint.** Find where $f' = 0$, then compute f there. The question asks for the value, not the x.

**Answer: B.** $f' = 2x - 8 = 0 \Leftrightarrow x = 4$; $f(4) = 16 - 32 + 30 =$ **14**. 4 is where the minimum is.

### 209 A maximum on an interval

What is the largest value of $f(x) = -x^2 + 6x$ on the interval $[0, 4]$?

**A.** 8
**B.** 0
**C.** 3
**D.** 16
**E.** None of the above

**Hint.** Compare f at the critical point inside the interval with f at both ends.

**Answer: E.** $f' = -2x + 6 = 0 \Leftrightarrow x = 3 \in [0, 4]$; $f(3) = 9$, $f(0) = 0$, $f(4) = 8$. Maximum **9**. 8 is the value at the end point.

> On this card the right answer — **9** — is not among A–D, so the answer is **E · None of the above**.

### 210 Ordering costs

A shop's yearly stock cost is $C(q) = q + \frac{400}{q}$, where q is the order size ($q > 0$). Which order size minimises it?

**A.** $q = 40$
**B.** $q = 200$
**C.** $q = 20$
**D.** $q = 400$
**E.** None of the above

**Hint.** $C'(q) = 1 - 400/q^2$. Set it to zero.

**Answer: C.** $q^2 = 400 \Leftrightarrow$ **$q = 20$** ($C = 40$). $C'' = 800/q^3 > 0$, so it is a minimum. 40 is the cost, not the order size.

### 211 The top of $x\,e^{-x}$

For $x \geq 0$, where does $f(x) = x\,e^{-x}$ reach its maximum?

**A.** $x = 1$
**B.** $x = e$
**C.** $x = 1/e$
**D.** $x = 0$
**E.** None of the above

**Hint.** $f'(x) = e^{-x}(1 - x)$. $e^{-x}$ is never 0.

**Answer: A.** $f' = 0 \Leftrightarrow x = 1$; $f' > 0$ before and $< 0$ after, so **$x = 1$** is the maximum ($f(1) = 1/e$). $1/e$ is the maximum VALUE, not where it occurs.

### 212 Smallest value on an interval

What is the smallest value of $f(x) = x^3 - 12x$ on the interval $[0, 3]$?

**A.** 16
**B.** 0
**C.** −16
**D.** −9
**E.** None of the above

**Hint.** Compare f at the critical points inside $[0, 3]$ with f at both ends.

**Answer: C.** $f' = 3x^2 - 12 = 0 \Leftrightarrow x = \pm 2$; only 2 lies in $[0, 3]$. $f(0) = 0$, $f(2) = -16$, $f(3) = -9$: minimum **−16**. −9 is the value at the end point; 2 is where the minimum is.

### 213 When the second-derivative test is silent

You know that $f'(2) = 0$ and $f''(2) = 0$. What can you conclude about $x = 2$?

**A.** A local maximum

**B.** An inflection point

**C.** Nothing yet: it may be a maximum, a minimum or neither

**D.** f is constant near 2

**E.** None of the above

**Hint.** Think of $x^4$, $-x^4$ and $x^3$ at $x = 0$ — all three have $f' = f'' = 0$ there.

**Answer: C.** $x^4$ has a minimum at 0, $-x^4$ a maximum, $x^3$ neither (an inflection point), and all three have $f'(0) = f''(0) = 0$. So the test gives **no conclusion**: look at the sign of $f'$ on each side instead.

### 214 Reading a sign chart

$f'(x) > 0$ on $]0, 3[$ and $f'(x) < 0$ on $]3, 6[$. What does f have at $x = 3$?

**A.** $f(3) = 0$

**B.** A local minimum

**C.** $f''(3) > 0$

**D.** An inflection point

**E.** None of the above

**Hint.** $f' > 0$: f rises. $f' < 0$: f falls. Rises, then falls …

**Answer: E.** f increases up to 3 and decreases after it: **a local maximum** at 3 (first-derivative test). Nothing says $f(3)$ itself is 0 — it is $f'(3)$ that changes sign.

> On this card the right answer — **A local maximum** — is not among A–D, so the answer is **E · None of the above**.

### 215 Where $e^x - 3x^2$ is concave up

On which interval is $f(x) = e^x - 3x^2$ concave up?

**A.** $]6, +\infty[$

**B.** $]\ln 6, +\infty[$

**C.** $]-\infty, \ln 6[$

**D.** $]0, +\infty[$

**E.** None of the above

**Hint.** Concave up where $f'' > 0$. $f' = e^x - 6x$.

**Answer: B.** $f'' = e^x - 6 > 0 \Leftrightarrow e^x > 6 \Leftrightarrow$ **$x > \ln 6 \approx 1.79$**. $]6, +\infty[$ forgets the ln; $]-\infty, \ln 6[$ is where f is concave down.